
Multimedia content transmission and distribution systems

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DAC – UPC

* Part of the material comes from other sources. See bibliography

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- Introduction
- Streaming
- Real time interactive application protocols
- Content distribution networks and multimedia contents
- New trends in MM protocols

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Multimedia support HTML

- HTML5 and the video element
 - https://www.w3schools.com/html/html5_video.asp
- HTML5 and the audio element
 - https://www.w3schools.com/html/html5_audio.asp
- HTML5 support in browsers
 - <https://html5test.com/index.html> (old version)
 - <https://html5test.co/>
- Web trends
 - https://w3techs.com/technologies/overview/web_server

Video support Edge over Windows 10

2017

2019

Video

33

video element	Yes ✓
Subtitles	Yes ✓
Audio track selection	Yes ✓
Video track selection	Yes ✓
Poster images	Yes ✓
Codec detection	Yes ✓
Video codecs	
MPEG-4 ASP support	Yes ✓
H.264 support	Yes ✓
H.265 support	Yes ✓
Ogg Theora support	No ✗
WebM with VP8 support	No ✗
WebM with VP9 support	No ✗

Video

33

video element	Yes ✓
Subtitles	Yes ✓
Audio track selection	Yes ✓
Video track selection	Yes ✓
Poster images	Yes ✓
Codec detection	Yes ✓
Video codecs	
MPEG-4 ASP support	Yes ✓
H.264 support	Yes ✓
H.265 support	Yes ✓
Ogg Theora support	Yes ✓
WebM with VP8 support	Yes ✓
WebM with VP9 support	Yes ✓

Audio support Edge over Windows 10

2017

Audio	27/30
audio element	Yes ✓
Loop audio	Yes ✓
Preload in the background	Yes ✓
Advanced	
Web Audio API	Yes ✓
Speech Recognition	No ✗
Speech Synthesis	Yes ✓
Audio codecs	
PCM audio support	Yes ✓
MP3 support	Yes ✓
AAC support	Yes ✓
Dolby Digital support	Yes ✓
Dolby Digital Plus support	Yes ✓
Ogg Vorbis support	No ✗
Ogg Opus support	No ✗
WebM with Vorbis support	No ✗
WebM with Opus support	No ✗

2019

Audio	27/30
audio element	Yes ✓
Loop audio	Yes ✓
Preload in the background	Yes ✓
Advanced	
Web Audio API	Yes ✓
Speech Recognition	No ✗
Speech Synthesis	Yes ✓
Audio codecs	
PCM audio support	Yes ✓
MP3 support	Yes ✓
AAC support	Yes ✓
Dolby Digital support	Yes ✓
Dolby Digital Plus support	Yes ✓
Ogg Vorbis support	Yes ✓
Ogg Opus support	Yes ✓
WebM with Vorbis support	Yes ✓
WebM with Opus support	Yes ✓

Streaming support Edge over Windows 10

2017

Streaming		5
Media Source extensions	Yes	✓
DRM support	Yes	✓
Adaptive bit rate		
Dynamic Adaptive Streaming / MPEG-DASH	Yes	✓
HTTP Live Streaming / HLS	Yes	✓
Codecs		
▼ Video codecs	Partial	○
MP4 with H.264 support	Yes	✓
MP4 with H.265 support	Yes	✓
TS with H.264 support	Yes	✓
TS with H.265 support	No	✗
WebM with VP8 support	No	✗
WebM with VP9 support	No	✗
▼ Audio codecs	Partial	○
MP4 with AAC support	Yes	✓
MP4 with Dolby Digital support	Yes	✓
MP4 with Dolby Digital Plus support	Yes	✓
TS with AAC support	Yes	✓
TS with Dolby Digital support	Yes	✓
TS with Dolby Digital Plus support	Yes	✓
WebM with Vorbis support	No	✗
WebM with Opus support	Yes	✓

2019

Streaming		5
Media Source extensions	Yes	✓
DRM support	Yes	✓
Adaptive bit rate		
Dynamic Adaptive Streaming / MPEG-DASH	Yes	✓
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MP4 with H.264 support	Yes	✓
MP4 with H.265 support	Yes	✓
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TS with H.265 support	No	✗
WebM with VP8 support	Yes	✓
WebM with VP9 support	No	✗
▼ Audio codecs	Partial	○
MP4 with AAC support	Yes	✓
MP4 with Dolby Digital support	Yes	✓
MP4 with Dolby Digital Plus support	Yes	✓
TS with AAC support	Yes	✓
TS with Dolby Digital support	No	✗
TS with Dolby Digital Plus support	No	✗
WebM with Vorbis support	Yes	✓
WebM with Opus support	Yes	✓

Video support Chrome Dev 77 over Windows 10 (2019)

Video	29/33
video element	Yes ✓
Subtitles	Yes ✓
Audio track selection	No ✗
Video track selection	No ✗
Poster images	Yes ✓
Codec detection	Yes ✓
Video codecs	
MPEG-4 ASP support	No ✗
H.264 support	Yes ✓
H.265 support	No ✗
Ogg Theora support	Yes ✓
WebM with VP8 support	Yes ✓
WebM with VP9 support	Yes ✓

Audio support Chrome Dev 77 over Windows 10 (2019)

Audio

29/30

audio element	Yes ✓
Loop audio	Yes ✓
Preload in the background	Yes ✓
Advanced	
Web Audio API	Yes ✓
Speech Recognition	Prefixed ✓
Speech Synthesis	Yes ✓
Audio codecs	
PCM audio support	Yes ✓
MP3 support	Yes ✓
AAC support	Yes ✓
Dolby Digital support	No ✗
Dolby Digital Plus support	No ✗
Ogg Vorbis support	Yes ✓
Ogg Opus support	Yes ✓
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WebM with Opus support	Yes ✓

- Check Chrome connections
- <chrome://net-export/>
- <chrome://flags/>

Streaming		5
Media Source extensions	Yes	✓
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MP4 with H.264 support	Yes	✓
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WebM with VP8 support	Yes	✓
WebM with VP9 support	Yes	✓
▼ Audio codecs	Partial	○
MP4 with AAC support	Yes	✓
MP4 with Dolby Digital support	No	✗
MP4 with Dolby Digital Plus support	No	✗
TS with AAC support	No	✗
TS with Dolby Digital support	No	✗
TS with Dolby Digital Plus support	No	✗
WebM with Vorbis support	Yes	✓
WebM with Opus support	Yes	✓

Video support Firefox 69 over Windows 10 (2019)

Video		29/33
video element	Yes	✓
Subtitles	Yes	✓
Audio track selection	No	✗
Video track selection	No	✗
Poster images	Yes	✓
Codec detection	Yes	✓
Video codecs		
MPEG-4 ASP support	No	✗
H.264 support	Yes	✓
H.265 support	No	✗
Ogg Theora support	Yes	✓
WebM with VP8 support	Yes	✓
WebM with VP9 support	Yes	✓

Audio support Firefox 69 over Windows 10 (2019)

Audio

27/30

audio element	Yes ✓
Loop audio	Yes ✓
Preload in the background	Yes ✓
Advanced	
Web Audio API	Yes ✓
Speech Recognition	No ✗
Speech Synthesis	Yes ✓
Audio codecs	
PCM audio support	Yes ✓
MP3 support	Yes ✓
AAC support	Yes ✓
Dolby Digital support	No ✗
Dolby Digital Plus support	No ✗
Ogg Vorbis support	Yes ✓
Ogg Opus support	Yes ✓
WebM with Vorbis support	Yes ✓
WebM with Opus support	Yes ✓

Streaming support Firefox 69 over Windows 10 (2019)

Streaming		5
Media Source extensions	Yes	✓
DRM support	Yes	✓
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TS with H.265 support	No	✗
WebM with VP8 support	Yes	✓
WebM with VP9 support	Yes	✓
▼ Audio codecs	Partial	○
MP4 with AAC support	Yes	✓
MP4 with Dolby Digital support	No	✗
MP4 with Dolby Digital Plus support	No	✗
TS with AAC support	No	✗
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WebM with Vorbis support	Yes	✓
WebM with Opus support	Yes	✓

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 - Multimedia applications examples
 - Content examples
- Streaming
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- New trends in MM protocols

Multimedia content transmission

- Protocols and techniques to work with multimedia contents in the Internet
- Exponential growth of **Audio and Video Streaming** in Internet in the last years
- Important growth of **M2M / IoT** traffic

Multimedia content transmission

- Some numbers:

<https://www.cisco.com/c/en/us/solutions/collateral/executive-perspectives/annual-internet-report/white-paper-c11-741490.html>

<https://www.statista.com/statistics/273018/number-of-internet-users-worldwide/>

<https://www.statista.com/statistics/617136/digital-population-worldwide/>

<https://www.cisco.com/c/en/us/solutions/executive-perspectives/annual-internet-report/air-highlights.html#>

Who transmits audio and video in the Internet?

Some examples

- Radio and TV
- On-line press
- Advertisements
- Spotify
- Youtube
- Zoom
- Teams
- Meet
- Whatsapp
- TikTok
- Instagram
- Twitch
- Discord
- ...

Multimedia (MM) applications

- Multimedia applications
 - Multimedia data transmission (basically audio and video)
 - Opposite requirements than traditional content applications (FTP, HTTP, email)
- But... Which ones?

MM applications requirements

- Sensitive to end-to-end delay
- Sensitive to delay variability
- Tolerate occasional losses
- In contrast with classic transfer files applications
 - **Opposite requirements**
 - Do not tolerate losses, but delay is not relevant if all data arrives

- Justification?

MM applications requirements: Justification

- **Delay:** If a packet arrives too late (sound or image that should have already been reproduced)
 - It is discarded, no sense to listen or view it
- **Delay variability:** If packets arrives with a lot of difference, maybe it cannot be properly reproduced
 - Minimize delay → *Buffering*
- **Losses:** High tolerance, if loss of an image or sound, the user can reproduce the rest of the content

Classical MM content types

- **Quicktime Movie**: Apple container for any kind of content. Support for dedicated streaming and HTTP
- **Real Media**: Real Network format for streaming and HTTP streaming
- **Windows Media**: Microsoft streaming and live transmissions

Classical MM content types

- **Quicktime Movie** - ^
container for
Sun

Originally dedicated Streaming,
now all converge to MPEG-
DASH

<https://www.wowza.com/blog/mpeg-dash-dynamic-adaptive-streaming-over-http>

transmissions

MM content types

- **MPEG:** Standard format defined by the Moving Picture Experts Group (MPEG).
 - Supports three information types: audio, video and streaming (synchronized audio and video)
- **Adobe Flash Video:** Video format created by Flash (.flv), supporting streaming.
 - Discontinued on 2021

<https://www.adobe.com/products/flashplayer/end-of-life-alternative.html#adobe-flash-player-eol-general-information>

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 - What is it?
 - Streaming types
- Real time interactive application protocols
- Content distribution networks and multimedia contents
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Streaming: What is it?

- “Video or audio content sent in compressed form over the Internet and played immediately” – techtarget.com
- “Multimedia that is constantly received by and presented to an end-user while being delivered” – wikipedia.org

Streaming

- The content I want to view / listen is not in my machine, it is played as it arrives
- Streaming types
 - Stored multimedia content
 - Live
 - Real time interactive

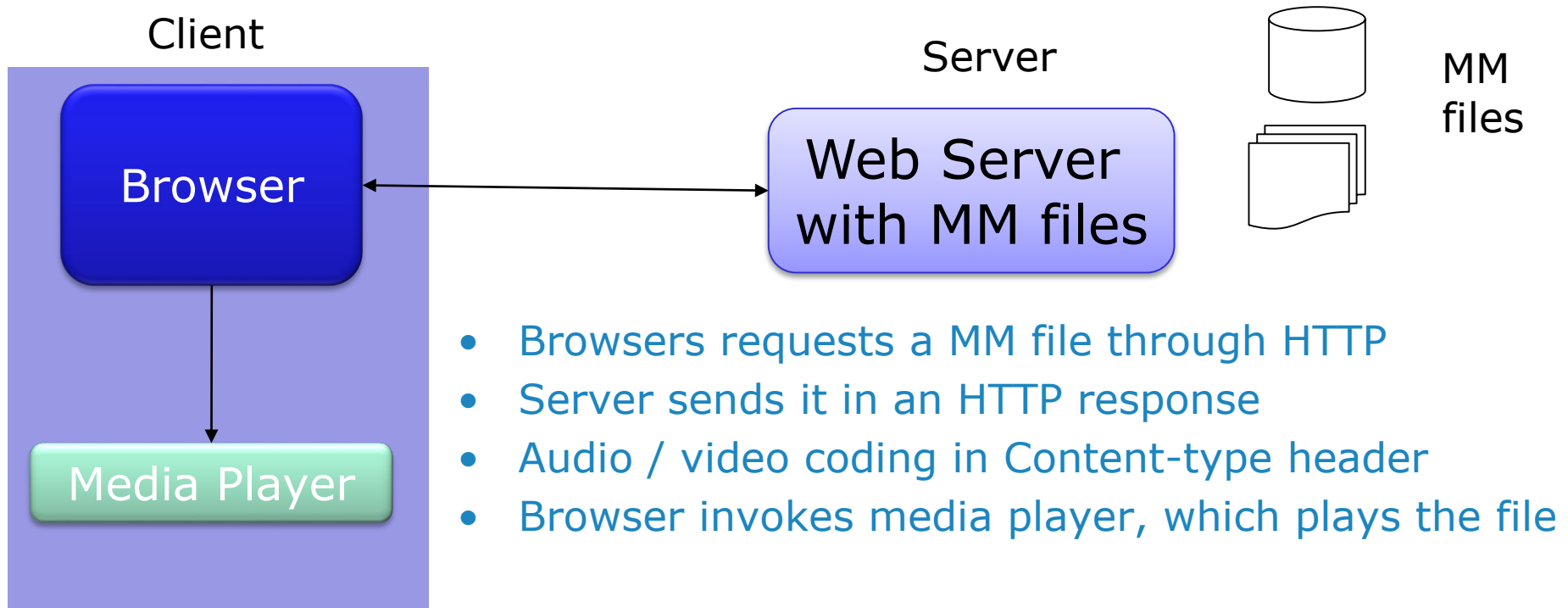
Streaming

- Multimedia
 - applications for transmitting and distributing data
 - content types that are transmitted
- Streaming techniques
 - Using web servers
 - RTSP protocol
 - MPEG-DASH

Streaming: Stored content

- The client asks for audio and video files to the server, receives and plays them
- Interactive
 - The user can control video reproduction (pause, resume, fast forward, rewind, etc.)
- Wait time
 - Between 1 and 10 seconds can pass since client requests content until it is displayed
- Example: Youtube

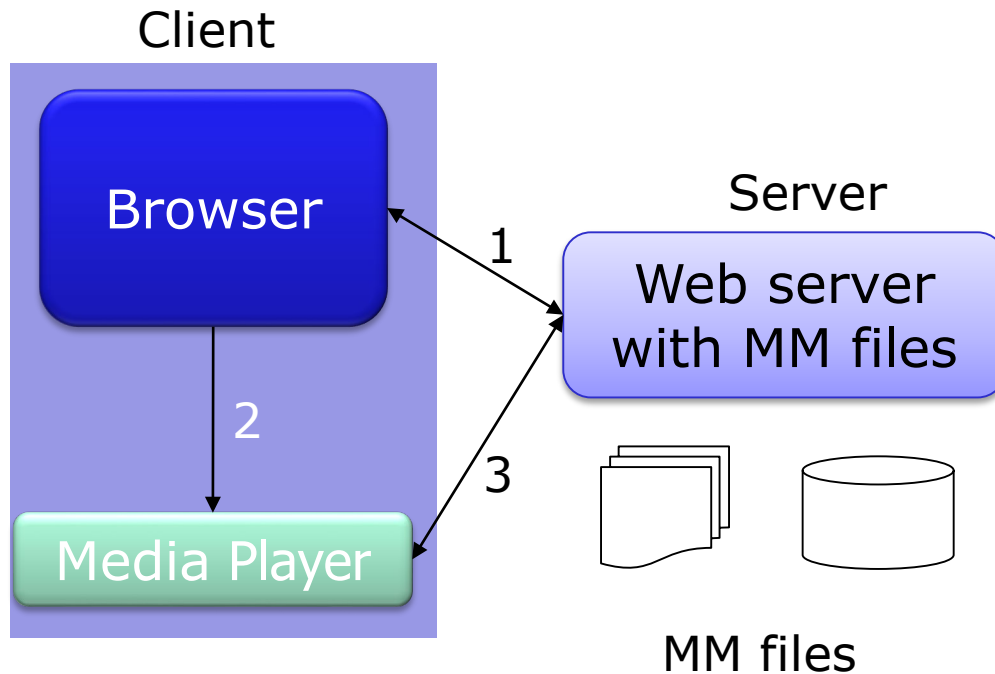
Progressive download from web server (I)



Problem

- Between media player and server there is an intermediary

Progressive download from web server (II)

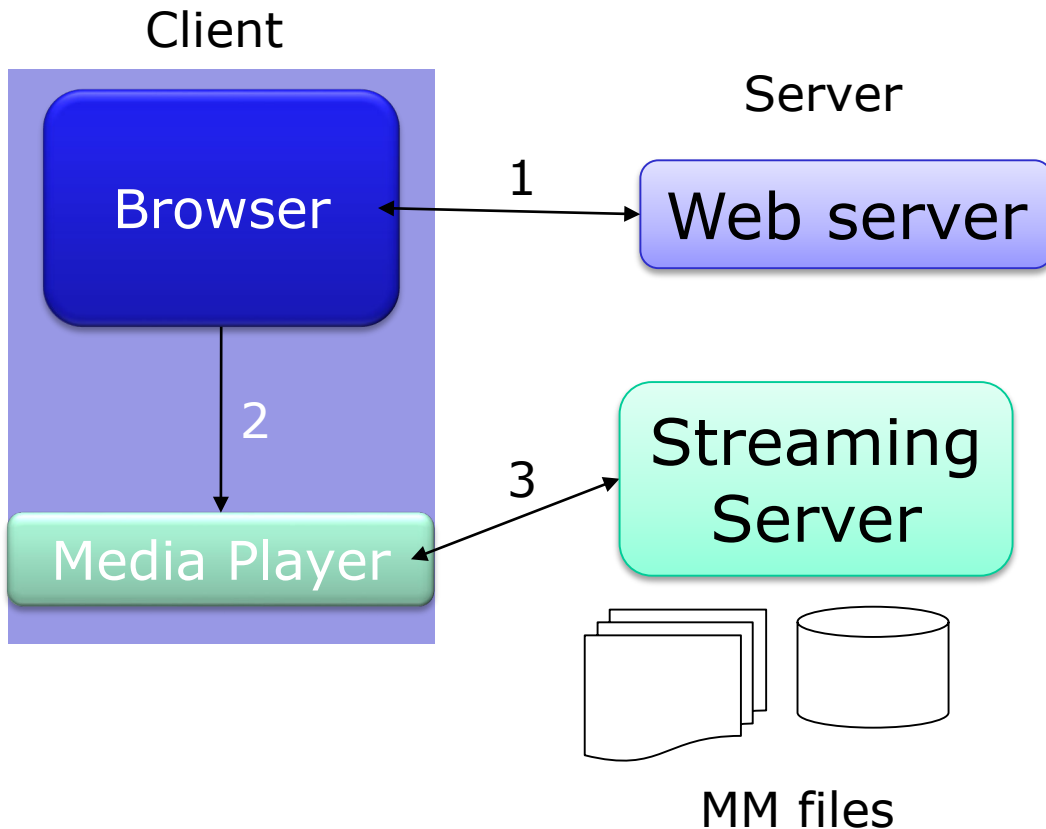


- Browser asks for a metafile file through HTTP (1)
- Server sends it in an HTTP response (1)
- Audio / video coding in Content-type header
- Browser passes the metafile file to a media player (2)
- Media player asks the web server the required file using HTTP (3)
- Server sends file to the media player, which plays it (3)

Considerations

- HTTP is not addressed to control playback (play, pause, ff, rnd, ...)
- Use of UDP for streaming?

Streaming with streaming server



- Browser asks for a file that describes the MM content (1)
- Browser passes the file to the media player (2)
- Media player establishes direct communication with the streaming server, which sends the MM files required

Considerations

- Communication between media player and web server does not have to be through HTTP
- UDP can be used

Live streaming

- Similar to TV and radio broadcasting, but it is done through the Internet
- Unidirectional and not interactive, only played or stopped
- Examples
 - Radio, on-line TV

Real time

- Audio / Video conference
- More restrictions in delay than in streaming
- Unidirectional as it is real time
 - Video: < 150 msec acceptable
 - Audio: < 150 msec good, < 400 msec acceptable

Streaming related protocols

- Real Time Streaming Protocol (RTSP) - 1998
- MPEG-DASH – 2014, subsequent revisions
- Commercial products
 - Google meet
 - Microsoft teams
 - Zoom
 - ...

Real Time Streaming Protocol (RTSP)

- Protocol to establish and control one or more multimedia synchronized streams
 - Audio and video
- Does **not** send multimedia data
 - Just control information to control data transmission
 - Remote control to multimedia data servers

Real Time Streaming Protocol (RTSP)

- Sessions maintained by the server
- Identifier per session. Not bound to a TCP connection
 - It is possible to open and close TCP connections in the same session
 - UDP can also be used
- It can send RTP streams
 - Multimedia streams

Real Time Streaming Protocol (RTSP)

- Similarities with HTTP 1.1
 - Share extension mechanisms
- Message format has:
 - request / response line
 - headers
 - message body

Real Time Streaming Protocol (RTSP)

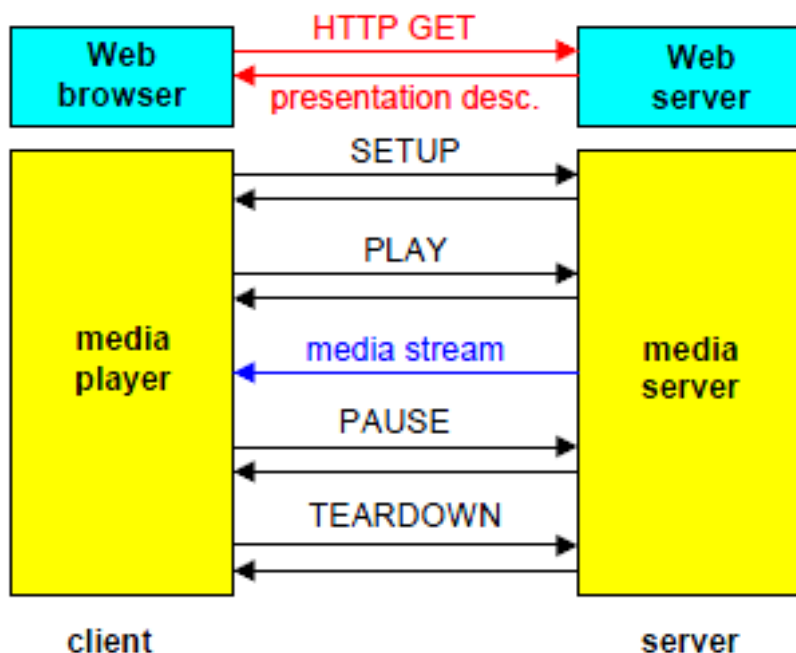
- Differences

RTSP	HTTP
Protocol Id rtsp://	Protocol Id http://
Maintains session state	Stateless
Client and server can make requests	In HTTP 1.1 this is not possible
Data is sent with another protocol	Data goes inside the message body
RTSP request URI always contains the complete URI, including the server	HTTP only has the resource name and the server name goes in a header

Real Time Streaming Protocol (RTSP)

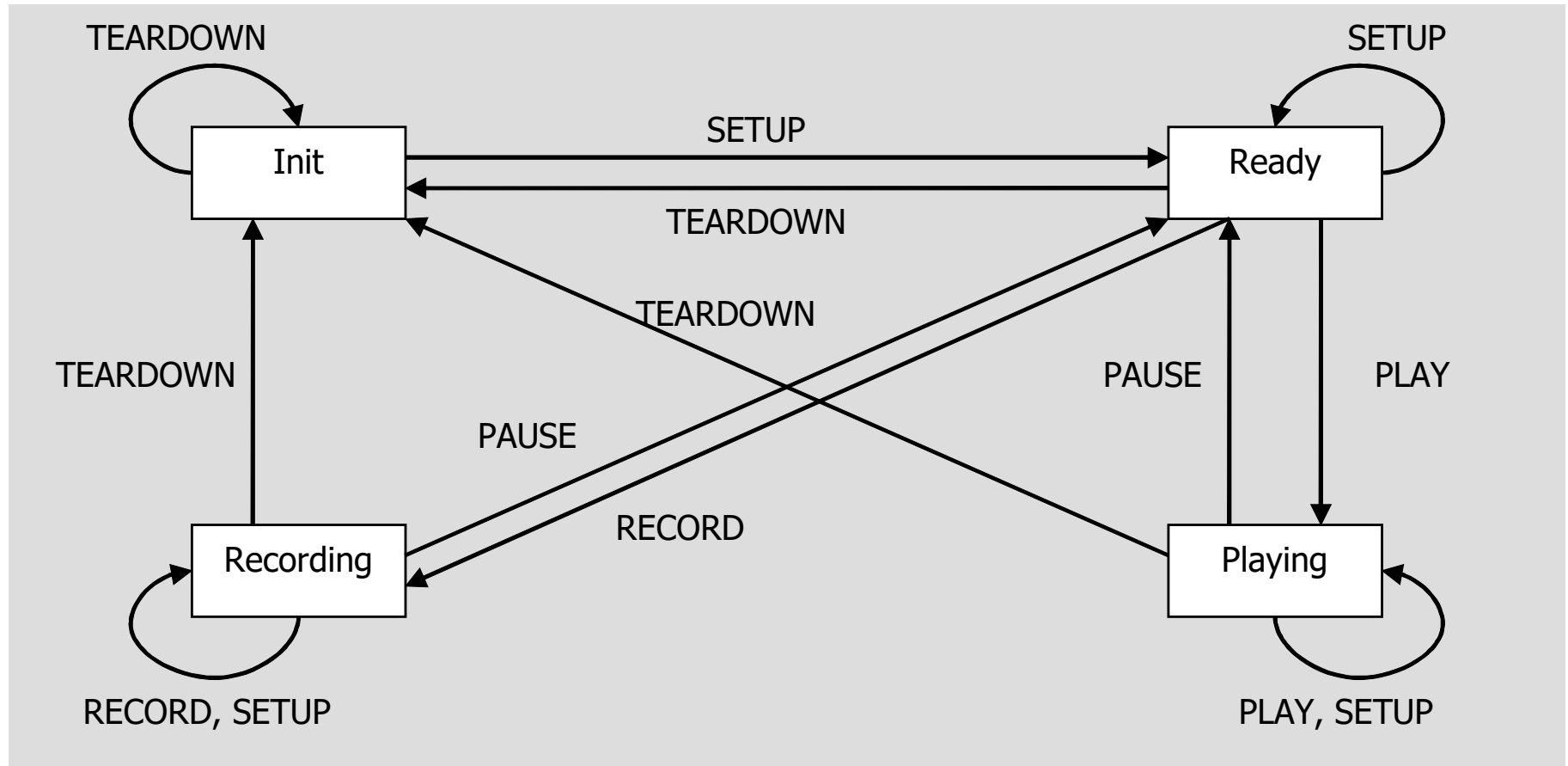
- **Operations**
 - Recover data from a multimedia content server
 - Invitation from a data server to a conference
 - Add content to a existing presentation

RTSP example



- ❑ Client obtains a description of the multimedia presentation, which can consist of several media streams.
- ❑ The browser invokes media player (helper application) based on the content type of the presentation description.
- ❑ Presentation description includes references to media streams, using the URL method `rtsp://`
- ❑ Player sends RTSP SETUP request; server sends RTSP SETUP response.
- ❑ Player sends RTSP PLAY request; server sends RTSP PLAY response.
- ❑ Media server pumps media stream.
- ❑ Player sends RTSP PAUSE request; server sends RTSP PAUSE response.
- ❑ Player sends RTSP TEARDOWN request; server sends RTSP TEARDOWN response.

RTSP client and server state diagram



Real Time Streaming Protocol (RTSP)

- **Methods**

- **OPTIONS**: Asks the types of request accepted by a server
- **SETUP**: The server reserves resources for a stream and starts an RTSP session
- **PLAY** and **RECORD**: Starts data transmission of a stream initialized with SETUP
- **PAUSE**: Stops transmission temporary without freeing resources from server
- **TEARDOWN**: Frees resources associated to the stream. The RTSP session is removed from server

<https://antmedia.io/rtsp-explained-what-is-rtsp-how-it-works/>

Real Time Streaming Protocol (RTSP)

- Some requests and responses examples

Real Time Streaming Protocol (RTSP)

- Client request

OPTIONS rtsp://servidor.multimedia.com:554

RTSP/1.0

Cseq: 1

- Server response

RTSP/1.0 200 OK

Cseq: 1

Public: DESCRIBE, SETUP, TEARDOWN, PLAY,
PAUSE

Real Time Streaming Protocol (RTSP)

- Client request

DESCRIBE

rtsp://servidor.multimedia.com:554/videos/video.html RTSP/1.0

Cseq: 2

- Server response

RTSP/1.0 200 OK

Cseq: 2

Content-type: application/sdp

Content-length: 210

<SDP Data...>

Real Time Streaming Protocol (RTSP)

- Client request

SETUP rtsp://

servidor.multimedia.com:554/videos/video.html

RTSP/1.0

Cseq: 3

Transport: rtp/udp;unicast;client_port=5067-5068

- Server response

RTSP/1.0 200 OK

Cseq: 3

Session: 12345678

Transport: rtp/udp;client_port=5067-
5068;server_port=6023-6024

Real Time Streaming Protocol (RTSP)

- Client request

PLAY rtsp://

servidor.multimedia.com:554/videos/video.mpg

RTSP/1.0

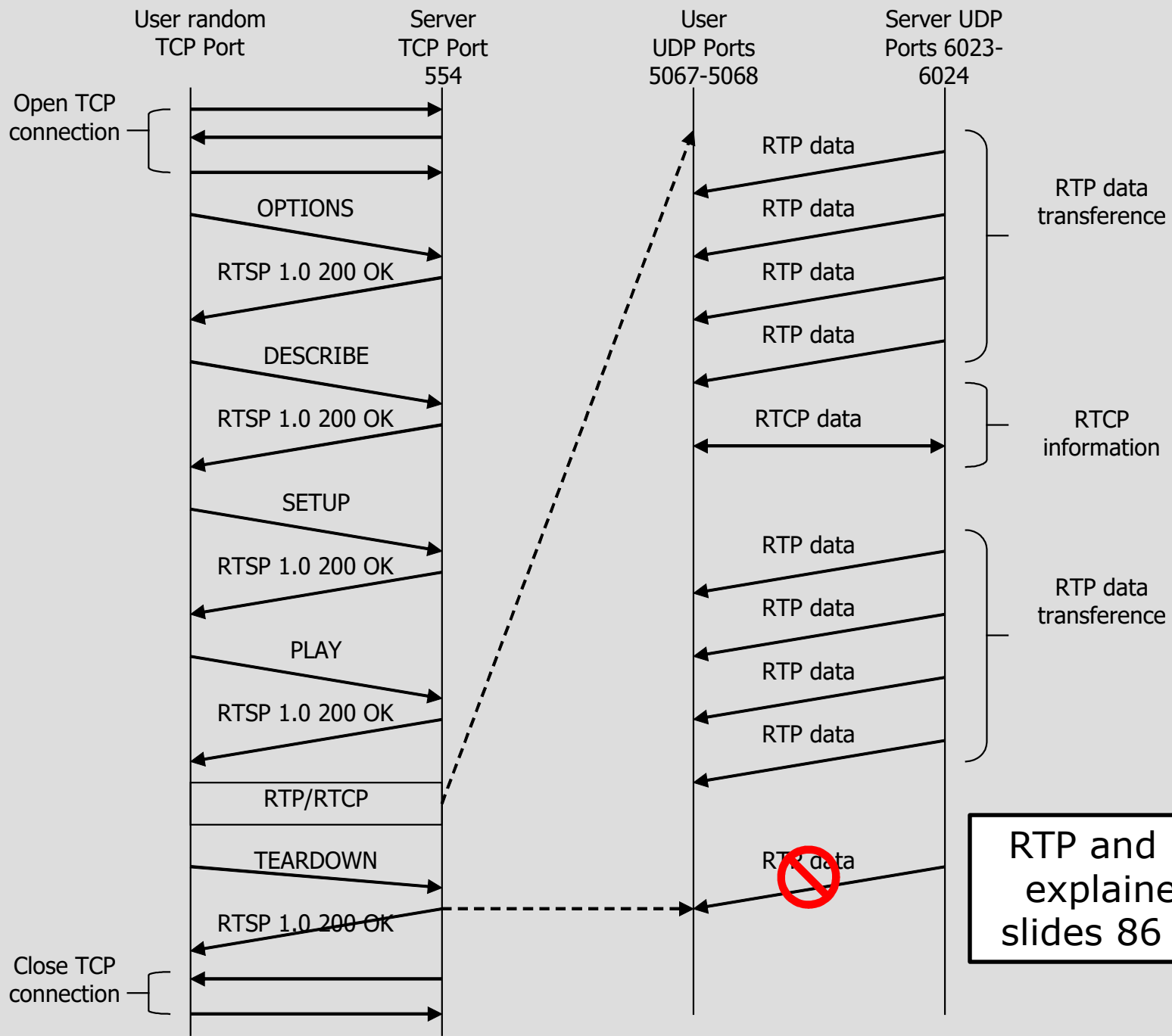
Cseq: 4

Session: 12345678

- Server response

RTSP/1.0 200 OK

Cseq: 4



RTP and RTCP explained in slides 86 to 89

Description file example

```
<title>Twister</title>
<session>
  <group language=en lipsync>
    <switch>
      <track type=audio
        e="PCMU/8000/1"
        src = "rtsp://audio.example.com/twister/audio.en/lofi">
      <track type=audio
        e="DVI4/16000/2" pt="90 DVI4/8000/1"
        src="rtsp://audio.example.com/twister/audio.en/hifi">
    </switch>
  <track type="video/jpeg"
    src="rtsp://video.example.com/twister/video">
</group>
</session>
```

RTSP session

- ❑ Each RTSP has a session identifier, which is chosen by the server.
- ❑ The client initiates the session with the SETUP request, and the server responds to the request with an identifier.
- ❑ The client repeats the session identifier for each request, until the client closes the session with the TEARDOWN request.
- ❑ RTSP port number is 554.
- ❑ RTSP can be sent over UDP or TCP. Each RTSP message can be sent over a separate TCP connection.

RTSP interchange

C: SETUP rtsp://audio.example.com/twister/audio RTSP/1.0
Transport: rtp/udp; compression; port=3056; mode=PLAY

S: RTSP/1.0 200 1 OK
Session 4231

C: PLAY rtsp://audio.example.com/twister/audio.en/lofi RTSP/1.0
Session: 4231
Range: npt=0-

C: PAUSE rtsp://audio.example.com/twister/audio.en/lofi RTSP/1.0
Session: 4231
Range: npt=37

C: TEARDOWN rtsp://audio.example.com/twister/audio.en/lofi RTSP/1.0
Session: 4231

S: 200 3 OK

OPTIONS rtsp://wowzaec2demo.streamlock.net:554/vod/mp4:BigBuckBunny_115k.mp4 RTSP/1.0
CSeq: 2
User-Agent: LibVLC/3.0.16 (LIVE555 Streaming Media v2016.11.28)

RTSP/1.0 200 OK
CSeq: 2
Server: Wowza Streaming Engine 4.8.20+1 build20220919162035
Cache-Control: no-cache
Public: DESCRIBE, SETUP, TEARDOWN, PLAY, PAUSE, OPTIONS, ANNOUNCE, RECORD, GET_PARAMETER
Supported: play.basic, con.persistent

DESCRIBE rtsp://wowzaec2demo.streamlock.net:554/vod/mp4:BigBuckBunny_115k.mp4 RTSP/1.0
CSeq: 3
User-Agent: LibVLC/3.0.16 (LIVE555 Streaming Media v2016.11.28)
Accept: application/sdp

RTSP/1.0 200 OK
CSeq: 3
Server: Wowza Streaming Engine 4.8.20+1 build20220919162035
Cache-Control: no-cache
Expires: Tue, 11 Oct 2022 08:14:08 UTC
Content-Length: 608
Content-Base: rtsp://wowzaec2demo.streamlock.net:554/vod/mp4:BigBuckBunny_115k.mp4/
Date: Tue, 11 Oct 2022 08:14:08 UTC
Content-Type: application/sdp
Session: 1056847574;timeout=60

v=0
o=- 1056847574 1056847574 IN IP4 34.227.104.115
s=BigBuckBunny_115k.mp4
c=IN IP4 34.227.104.115
t=0 0

ACAAGxAAAADAEEAAAaWdxQplgA==,a0vssiw=

a=cliprect:0,0,160,240

a=framesize:97 240-160

a=framerate:24.0

a=control:trackID=2

SETUP rtsp://wowzaec2demo.streamlock.net:554/vod/mp4:BigBuckBunny_115k.mp4/trackID=1 RTSP/1.0

CSeq: 4

User-Agent: LibVLC/3.0.16 (LIVE555 Streaming Media v2016.11.28)

Transport: RTP/AVP;unicast;client_port=65330-65331

RTSP/1.0 200 OK

CSeq: 4

Server: Wowza Streaming Engine 4.8.20+1 build20220919162035

Cache-Control: no-cache

Expires: Tue, 11 Oct 2022 08:14:08 UTC

Transport: RTP/AVP;unicast;client_port=65330-65331;source=34.227.104.115;server_port=7084-7085;ssrc=240163D7

Date: Tue, 11 Oct 2022 08:14:08 UTC

Session: 1056847574;timeout=60

SETUP rtsp://wowzaec2demo.streamlock.net:554/vod/mp4:BigBuckBunny_115k.mp4/trackID=2 RTSP/1.0

CSeq: 5

User-Agent: LibVLC/3.0.16 (LIVE555 Streaming Media v2016.11.28)

Transport: RTP/AVP;unicast;client_port=65332-65333

Session: 1056847574

RTSP/1.0 200 OK

CSeq: 5

Server: Wowza Streaming Engine 4.8.20+1 build20220919162035

Cache-Control: no-cache

Expires: Tue, 11 Oct 2022 08:14:08 UTC

Transport: RTP/AVP;unicast;client_port=65332-65333;source=34.227.104.115;server_port=7096-7097;ssrc=68B41867

Date: Tue, 11 Oct 2022 08:14:08 UTC

Session: 1056847574;timeout=60

Session: 1056847574;timeout=60

PLAY rtsp://wowzaec2demo.streamlock.net:554/vod/mp4:BigBuckBunny_115k.mp4/ RTSP/1.0

CSeq: 6

User-Agent: LibVLC/3.0.16 (LIVE555 Streaming Media v2016.11.28)

Session: 1056847574

Range: npt=0.000-

RTSP/1.0 200 OK

RTP-Info: url=rtsp://wowzaec2demo.streamlock.net:554/vod/mp4:BigBuckBunny_115k.mp4/

trackID=1;seq=1;rtptime=0,url=rtsp://wowzaec2demo.streamlock.net:554/vod/mp4:BigBuckBunny_115k.mp4/

trackID=2;seq=1;rtptime=0

CSeq: 6

Server: Wowza Streaming Engine 4.8.20+1 build20220919162035

Cache-Control: no-cache

Range: npt=0.0-634.625

Session: 1056847574;timeout=60

PAUSE rtsp://wowzaec2demo.streamlock.net:554/vod/mp4:BigBuckBunny_115k.mp4/ RTSP/1.0

CSeq: 7

User-Agent: LibVLC/3.0.16 (LIVE555 Streaming Media v2016.11.28)

Session: 1056847574

RTSP/1.0 200 OK

CSeq: 7

Server: Wowza Streaming Engine 4.8.20+1 build20220919162035

Cache-Control: no-cache

Session: 1056847574;timeout=60

rtsp://62.109.19.230:554/iloveyou
http://streams.videolan.org/

Revisar web rtsp

Wireshark · Seguir flujo UDP (udp.stream eq 7) · Wi-Fi

— □

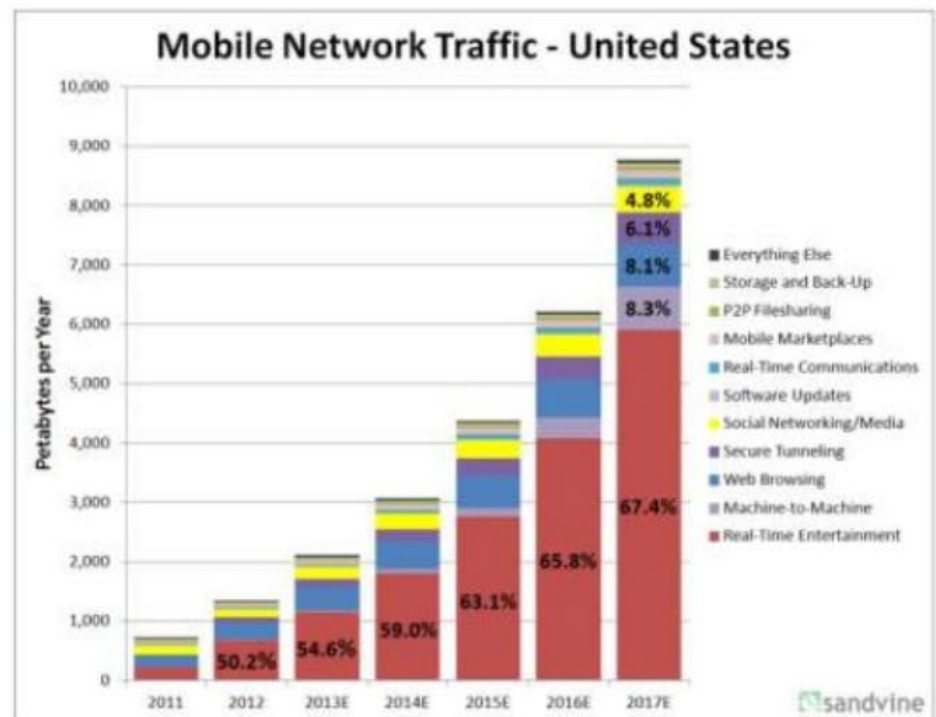
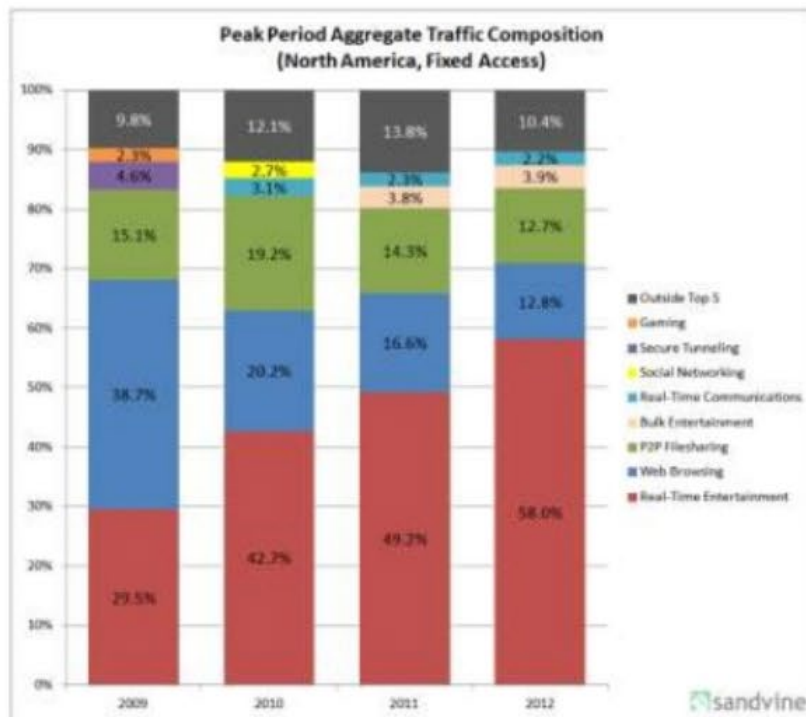
```
.....h..g...gd....C.o....l@....@.....
e...h...,.....E....H.,. .#.x264 - core 163 - H.264/MPEG-4 AVC codec - Copyleft 2003-2021 - http://
www.videolan.org/x264.html - options: cabac=1 ref=3 deblock=1:0:0 analyse=0x3:0x113 me=hex subme=7 psy=1
psy_rd=1.00:0.00 mixed_ref=1 me_range=16 chroma_me=1 trellis=1 8x8dct=1 cqm=0 deadzone=21,11 fast_pskip=0
chroma_qp_offset=-2 threads=5 lookahead_threads=1 sliced_threads=0 nr=0 decimate=1 interlaced=0
bluray_compat=0 constrained_intra=0 bframes=3 b_pyramid=2 b_adapt=1 b_bias=0 direct=1 weightb=1 open_gop=0
weightp=2 keyint=250 keyint_min=24 scenecut=40 intra_refresh=0 rc_lookahead=40 rc=abr mbtree=1 bitrate=84
ratetol=1.0 qcomp=0.60 qpmin=0 qpmax=69 qpstep=4 ip_ratio=1.40 aq=1:1.00....e...
5....].....*}.x.....E.....Zn....B.....3.]#..r    O.....m..
....7J.....<...J.N..a..<j...M..@.p..T-Ss....*.`4st.I.^9..aGw...M..z.....X..).W./Y{...^D|....
\sF.....8..U.YQ....Cw.c2..gVp....yu.F.....`....yM.....,..o*.l....W..R.....v....0....
....B....Ze..x.....:..h..gA."c.r..9...#..9<
8x.}..|..`.....P..[...0G..H.....E*.....q.....mr..Q...K$...N.....aE]..j>..n..J...N.{4M.....b!
1.&..Y&...-p..6.....L^.....}&q).....Y0.....+..h..g..Ay...P.h>..E.....I h..gA.CI..C ..
02@s...!.....;@.I...^.....e#.T..W...`5.?9.
jP-..Uo.....z..7.9..A
x1/..].
rA....s]'...kJ.....f.{YpY....P..4.tS..y...j)(.-}.u...E.....V.?1.f.9.-.iT.....+.3....<8.AE$..
1xe4i.s#.....u..... 6...:'5.=3}@.....(.].....T....Di4.KsG.= ..E..... ..e...x.!...S.
j/.....:..... ..eI.".
.K.%..a....b.Z.|.....,y....1.o.ZJv..^p....{'7..8}~....3C....3.....h.....f.h..gA.eK..C.SA0 ..
~....#..S...(+_?|yao.... ..|..F..8.....D...;:j...-5...|.....R.17...
\.]#.*.....:..@..i...I.j;iS*....k/....1z.-..u`/'viN.Y.....:b....#..~...tX...Pb..)...)
6..UK.....a...B...YX:4?..?pS...d....X..Eo#...Q.w...c7RU.../'..Z.6.....[h.Jb....
{.....k.g...D.....*....A....Pk4....e..R).n].u.&..IFk.).3..qce..
[*...]u.&..m.'..ZU..9Y..J{\!z.:U...,v...3...,9.Z[,v..`..t....=.....M..sq.....-.../...lmQ..19.C.x9..<79D.
03~..8..r..=. .#9.....{9....uB.P....)....s..=Y.....=.....K.1.E...q.Zo5....S.-.,...`I-.C.%....$.L4..H}mm.
$S?.....{$7.R..OVh6..P$'.....
...%o~#.>...h.K.U.y....x....7..et..G..`...L..1F}..9*...bwJ^...I^p..1
pE...2.Rm...8...k...M...d.N...*...>..h..H...z+....3M.TWf0!.4..@~...j<....{.l=U.n.....y..3..
.E.A.dC.....-S..o..t.....a.....l...jM.P.L76.sJ..`.....b.....2.....n.....I.M '.h...)Hy....`.l
```


MPEG-DASH

- MPEG Dynamic Adaptive Streaming over HTTP
- Why it appears?

VIDEO PREDOMINANT ON THE INTERNET

- Real-time video is **more than 50%** of the traffic at peak periods
- Mobile traffic is growing exponentially, all delivered over the top (OTT)



... AND HETEROGENEOUS DEVICES

Desktop/Laptop

Mobile

Living Room



... AND HETEROGENEOUS DEVICES

News from 2022/10/10

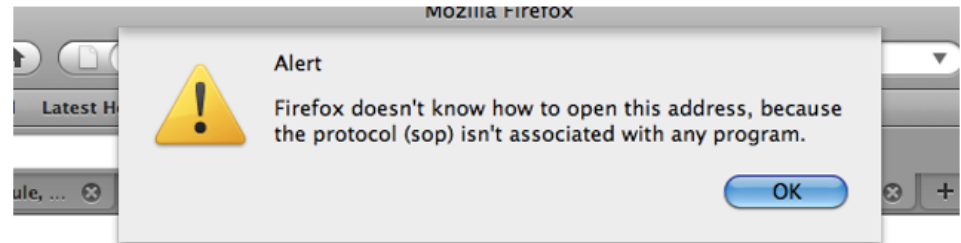
<https://techcrunch.com/2022/10/10/samsung-seeks-smart-tv-growth-with-first-tizen-os-licensing-deals/>



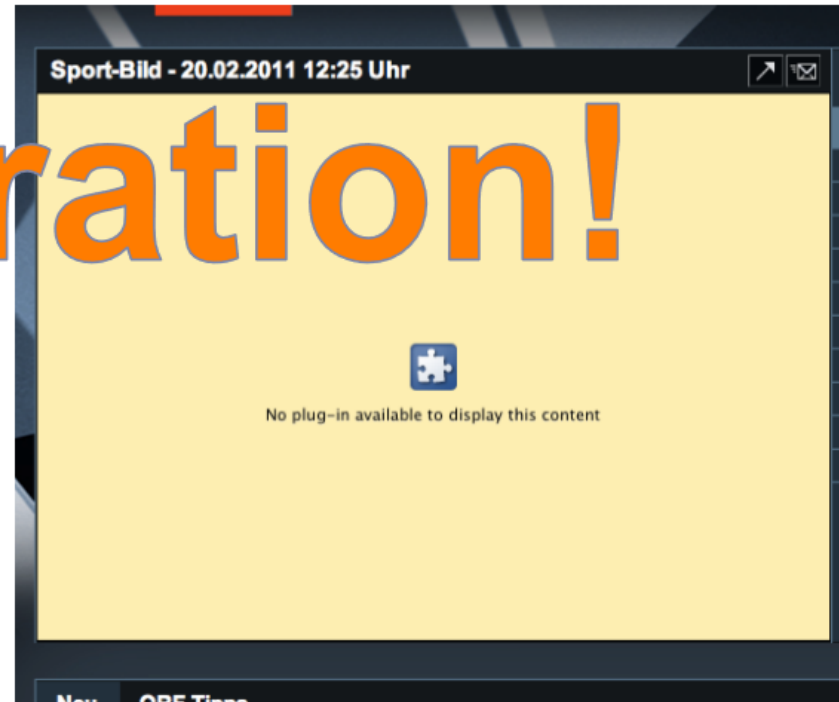
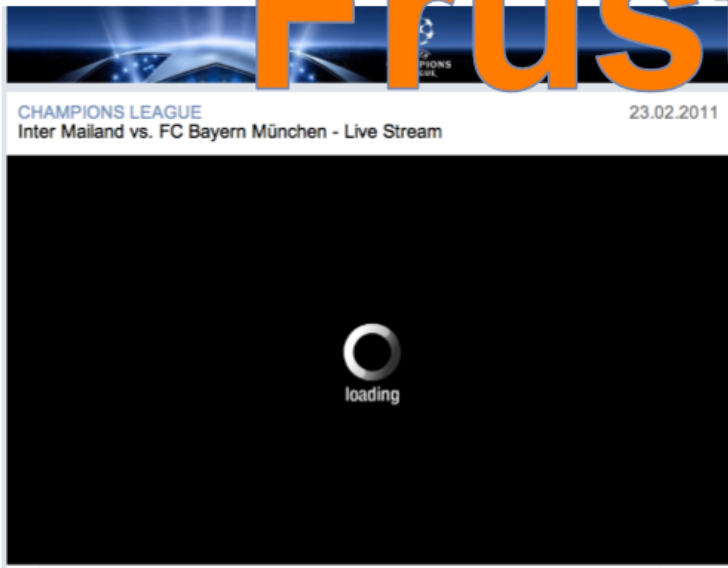
<https://www.ietf.org/proceedings/87/slides/slides-87-icnrg-5.pdf>

Before DASH...

Internazionale vs Bayern Munchen



Frustration!

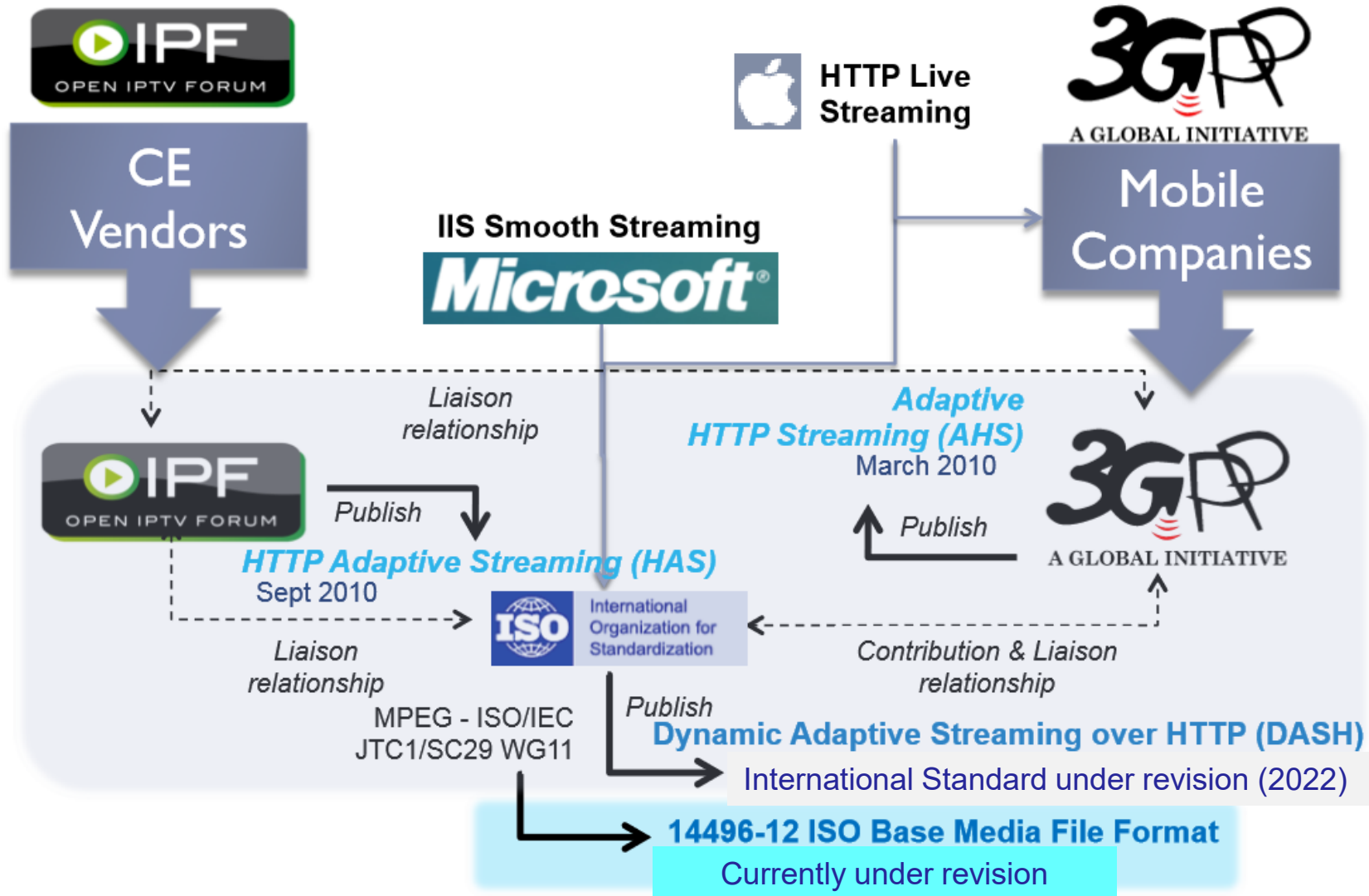


User Frustration in Internet Video

- ▶ **Video not accessible**
 - ▶ Behind a firewall
 - ▶ Plugin not available
 - ▶ Bandwidth not sufficient
 - ▶ Wrong/non-trusted device
 - ▶ Wrong format
- ▶ **Fragmentation**
 - ▶ Devices
 - ▶ Content Formats
 - ▶ DRM
- ▶ **Low quality of experience**
 - ▶ Long start-up delay
 - ▶ Frequent Rebuffering
 - ▶ Low playback quality
 - ▶ No lip-sync
 - ▶ No DVD quality (language, subtitle)
- ▶ **Expensive**
 - ▶ Sucks my bandwidth
 - ▶ Need a dedicated device
 - ▶ Other costs ...



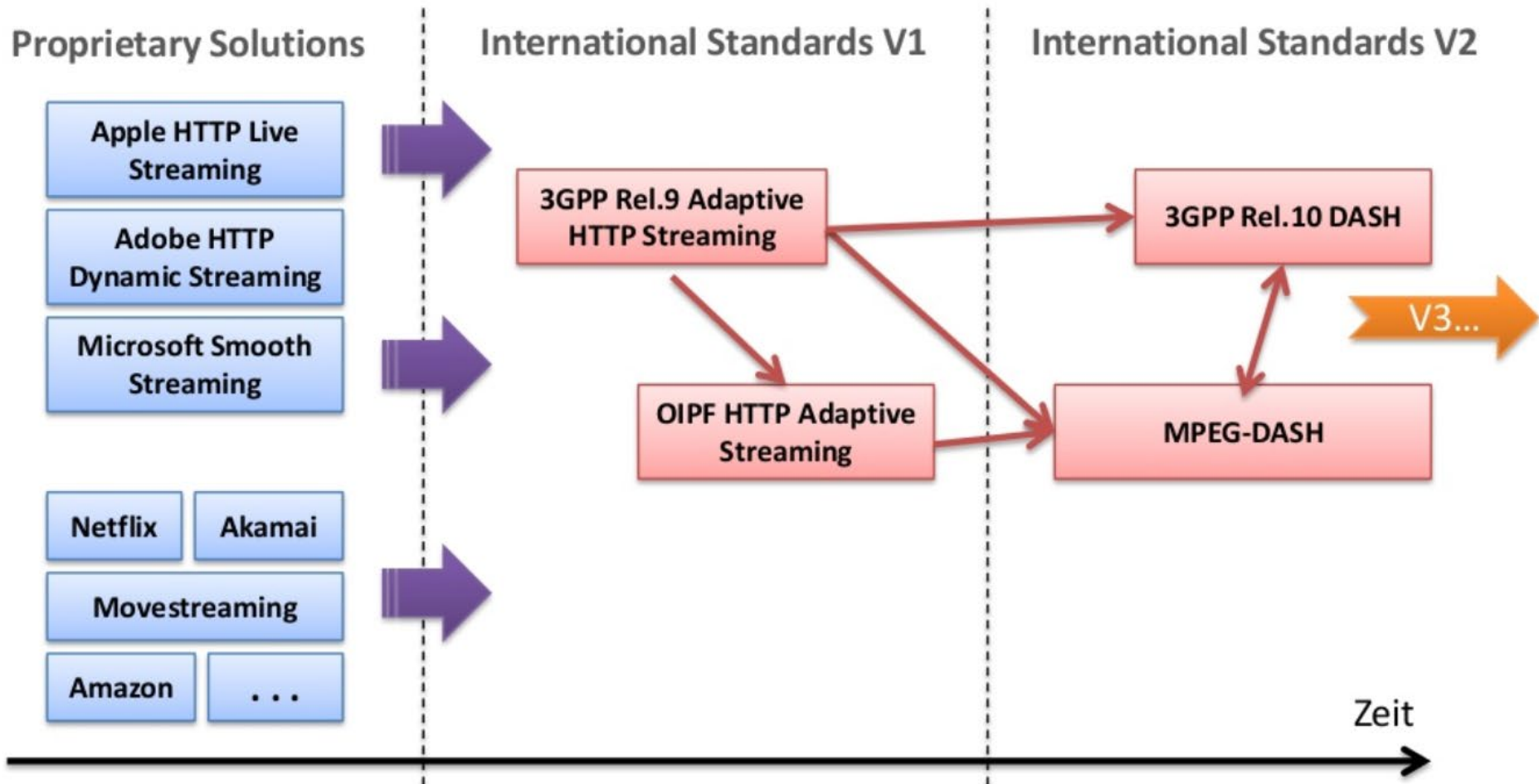
DASH: Standardization History and Status



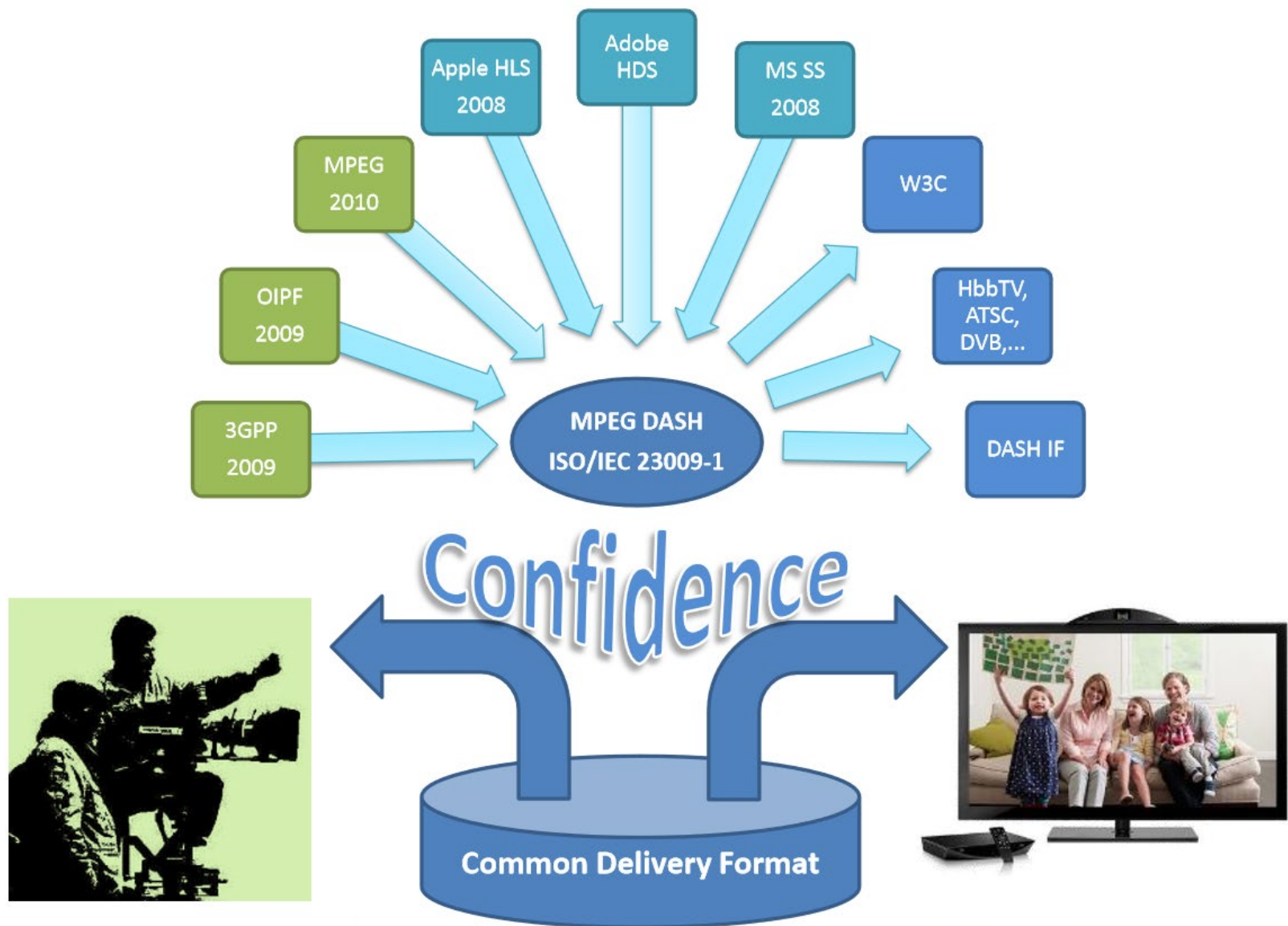
MPEG-DASH (ISO/IEC 23009-1:2022)

- Allows streaming over HTTP
- Interoperability between commercial products (Apple, Microsoft, Adobe → Standard)
- Solve problems from other MM content transmission protocols, like RTP
 - Firewalls
 - CDN's
 - Scalability (it is required one streaming session per client)
- <https://www.iso.org/standard/83314.html>
 - New version: <https://www.iso.org/standard/89027.html>
- DASH Industry forum <https://dashif.org/members/>

DYNAMIC ADAPTIVE STREAMING OVER HTTP (DASH)



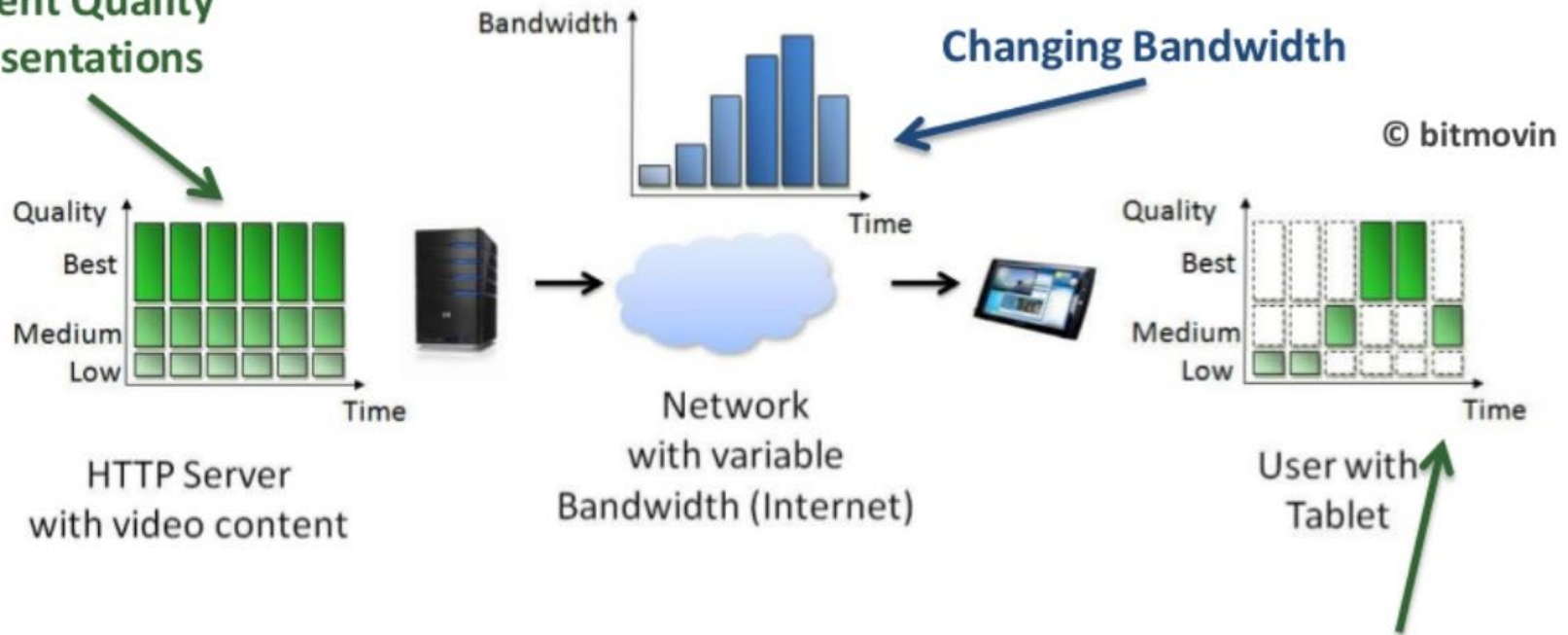
Source: <https://es.slideshare.net/slederer/mpegdash-open-source-tools-and-cloud-services>



<https://eclass.uoa.gr/modules/document/file.php/D476/Previous%20courses/2017-2018/An%20introduction%20to%20MPEG-DASH.pdf>

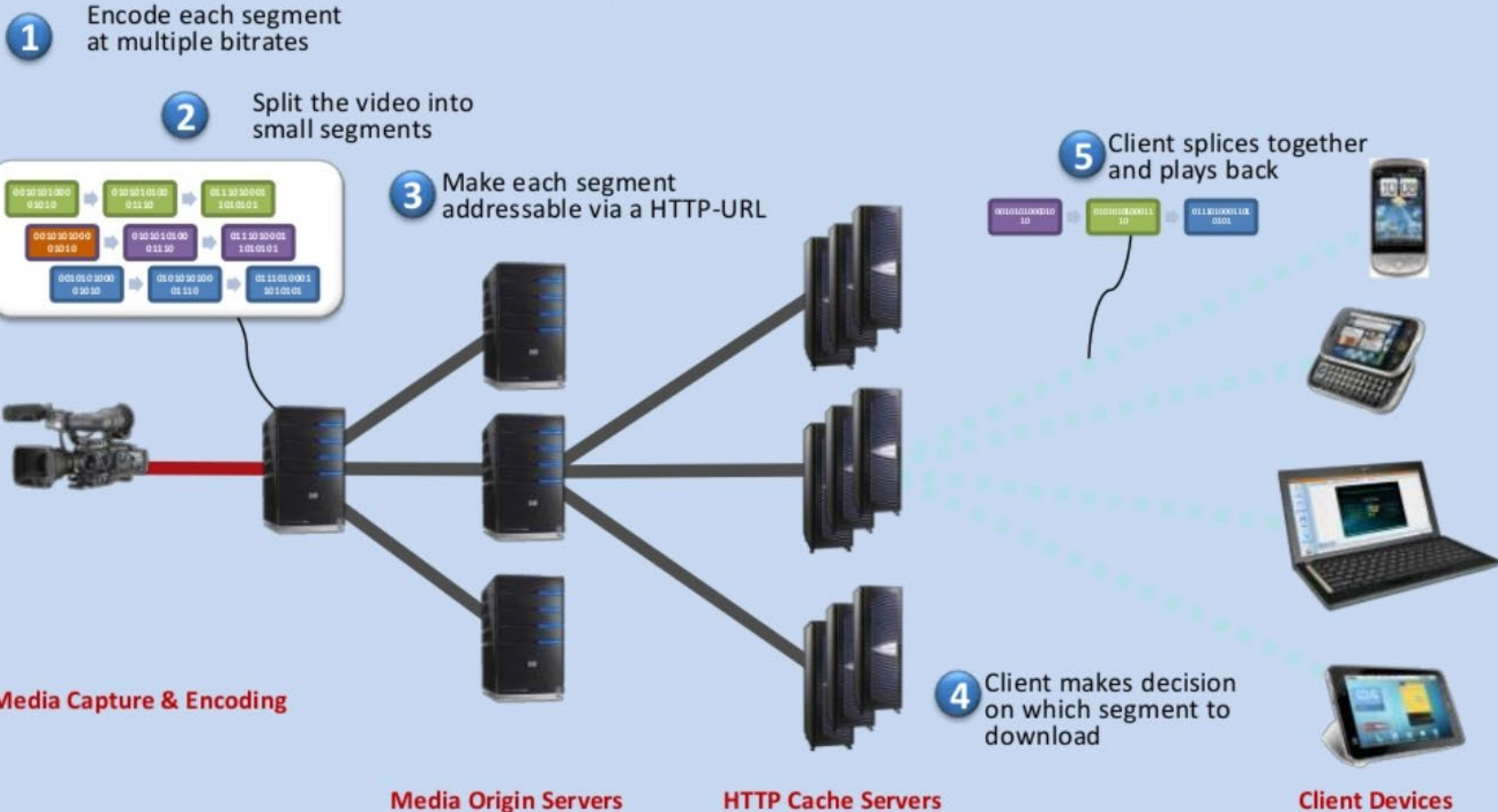
DYNAMIC ADAPTIVE STREAMING OVER HTTP

Different Quality Representations



- **Dynamic adaption** to the network conditions
- Usage of existing and cheap **Internet (HTTP) Infrastructure**
- Streaming-Logic is located at the **Client**
- **Flexible** and **scalable**

HTTP Adaptive Streaming



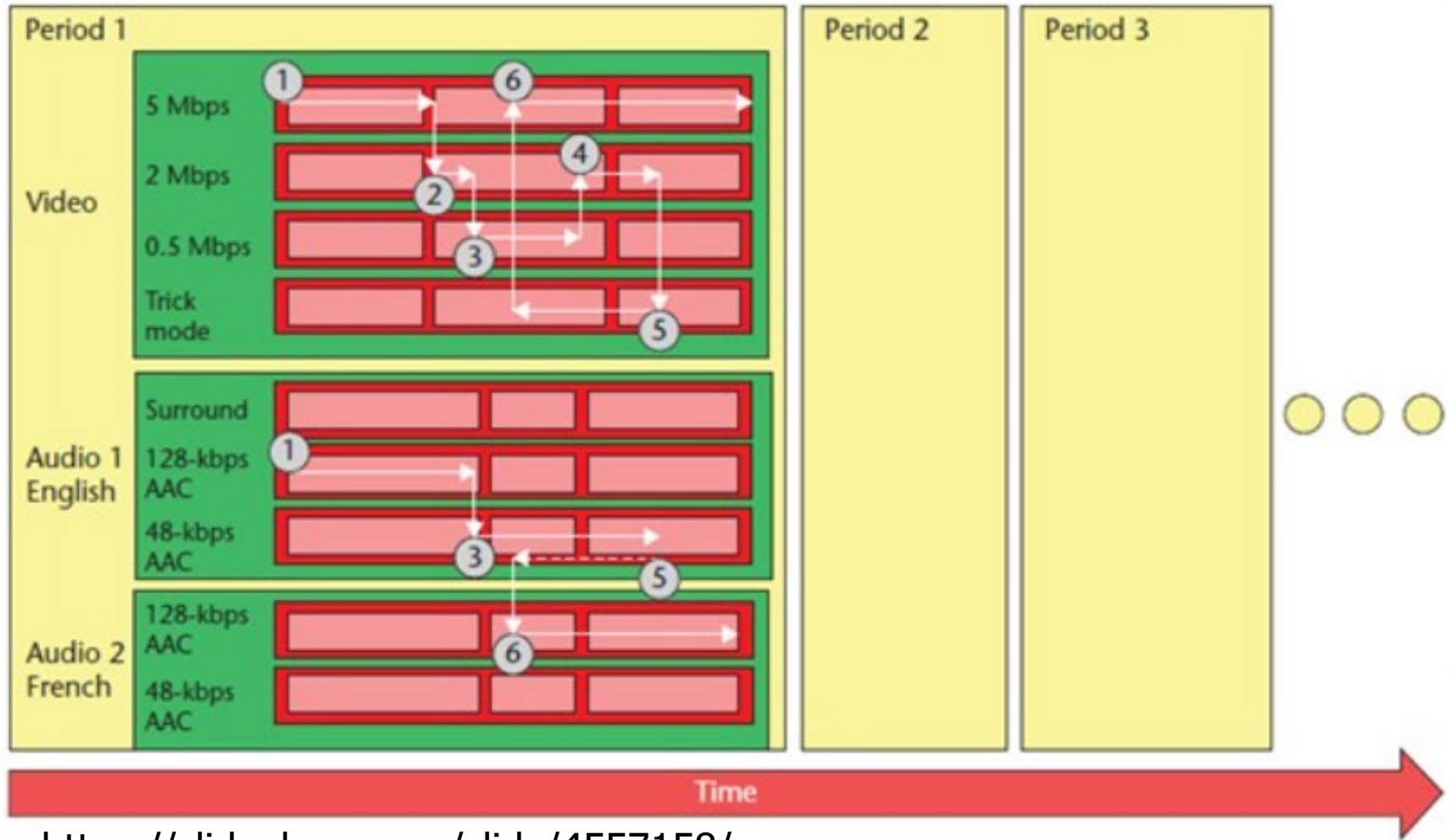
Advantages MPEG-DASH

- Internet has evolved to HTTP
 - Advanced support for data transmission using it
 - CDN's allow caching of content in local servers to reduce long distance traffic
 - HTTP goes smoothly through firewalls and proxies
 - No session with server is required
 - Client controls session
 - The number of clients is not an additional cost
-

MPEG-DASH features

- Supported stream types
 - ISO Base media file format (full, on-demand, live profiles)
 - MPEG-2 Transport Stream (TS) (main, simple profiles)
- Multimedia Presentation Description (MPD)
 - Describes the streamed MM content
 - XML format

Adaptive Streaming example – DVD

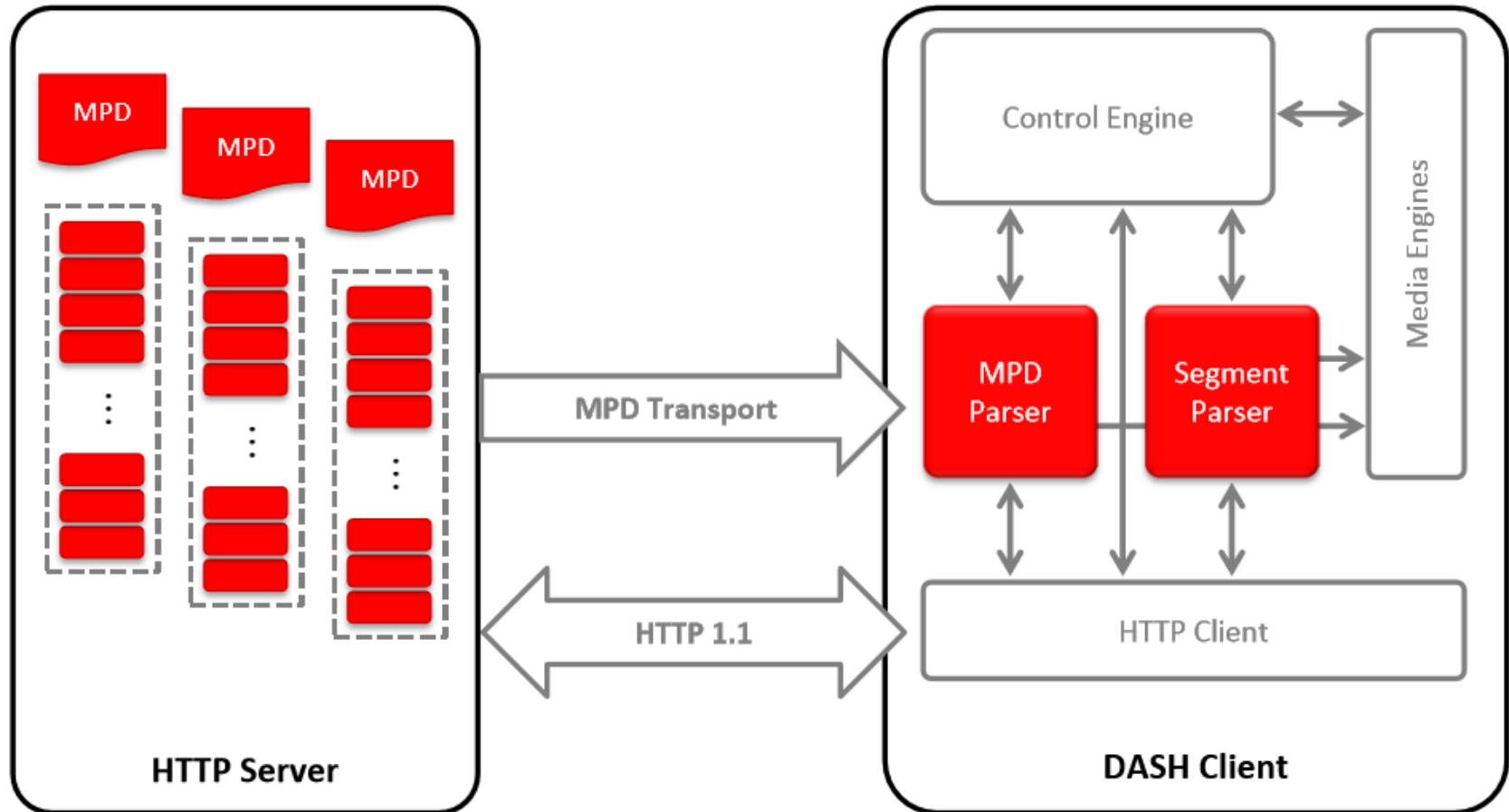


<https://slideplayer.com/slide/4557158/>

Adaptive streaming example explained

1. Stream of the first audio and video segment at 5M, bandwidth is not enough, it is reduced to 2
2. Stream of the second segment audio and video at 2M, bandwidth is not enough, it is reduced to 0,5M and 48K
3. Stream of the third segment at 0,5M, increased to 2M
4. User stops and rewinds
5. User uses trickmode, plays backwards without sound
6. User plays video again (5M) with French audio

Streaming with DASH

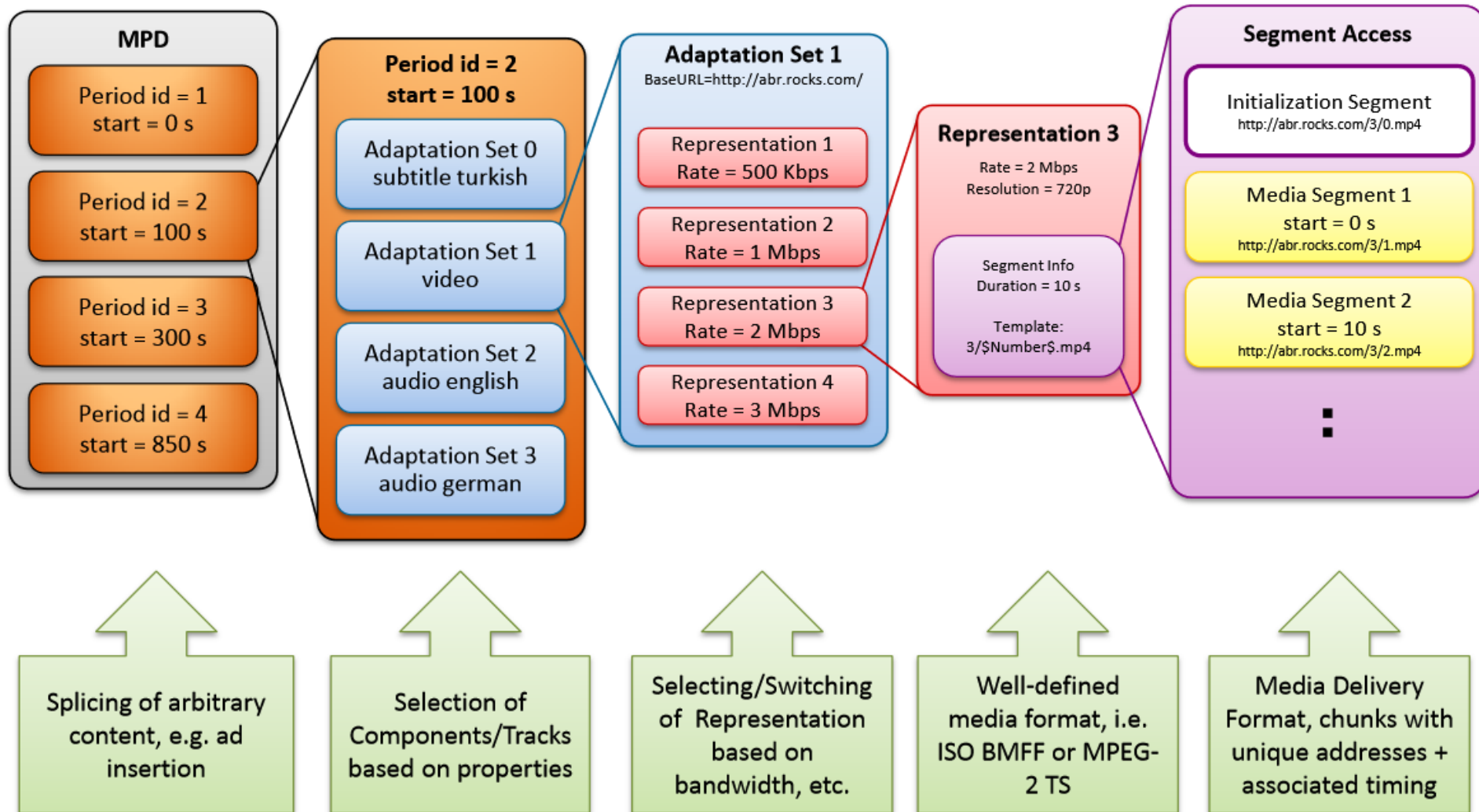


Source: <https://eclass.uoa.gr/modules/document/file.php/D476/Previous%20courses/2017-2018/An%20introduction%20to%20MPEG-DASH.pdf>

Architecture MPEG-DASH

- Server is a “classic” web server
- Client downloads two types of files
 - MPD: Multimedia Presentation Description
 - Files with the MM content described in the MPD
 - Content can be in different qualities
 - Content can be in different languages
- The client manages content download, calculating bandwidth constantly

MPD structure



MPD structure

- Hierarchical model
 - 1st level: Period, part of a program
 - 2nd level: Adaptation Set, provides different versions of the same content, with different coding alternatives
 - 3rd level: Representation, several representations of the same component
 - 4th level: Segment, media chunk streams that follow a temporal sequence

DASH MPD Schema (1/5)

<https://github.com/MPEGGroup/DASHSchema>

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema targetNamespace="urn:mpeg:DASH:schema:MPD:2011"
  attributeFormDefault="unqualified"
  elementFormDefault="qualified"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  xmlns:xlink="http://www.w3.org/1999/xlink"
  xmlns="urn:mpeg:DASH:schema:MPD:2011">

  <xs:import namespace="http://www.w3.org/1999/xlink"
    schemaLocation="xlink.xsd"/>

  <xs:annotation>
    <xs:appinfo>Media Presentation Description</xs:appinfo>
    <xs:documentation xml:lang="en">
      This Schema defines the Media Presentation Description for MPEG-DASH.
    </xs:documentation>
  </xs:annotation>

  <!-- MPD: main element -->
  <xs:element name="MPD" type="MPDtype"/>

  <!-- MPD Type -->
```

DASH MPD Schema (2/5)

<!-- MPD Type -->

```
<xs:complexType name="MPDtype">
  <xs:sequence>
    <xs:element name="ProgramInformation" type="ProgramInformationType" minOccurs="0" maxOccurs="unbounded" />
    <xs:element name="BaseURL" type="BaseURLType" minOccurs="0" maxOccurs="unbounded" />
    <xs:element name="Location" type="xs:anyURI" minOccurs="0" maxOccurs="unbounded" />
    <xs:element name="Period" type="PeriodType" maxOccurs="unbounded" />
    <xs:element name="Metrics" type="MetricsType" minOccurs="0" maxOccurs="unbounded" />
    <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded" />
  </xs:sequence>
  <xs:attribute name="id" type="xs:string" />
  <xs:attribute name="profiles" type="xs:string" use="required" />
  <xs:attribute name="type" type="PresentationType" default="static" />
  <xs:attribute name="availabilityStartTime" type="xs:dateTime" />
  <xs:attribute name="availabilityEndTime" type="xs:dateTime" />
  <xs:attribute name="mediaPresentationDuration" type="xs:duration" />
  <xs:attribute name="minimumUpdatePeriod" type="xs:duration" />
  <xs:attribute name="minBufferTime" type="xs:duration" use="required" />
  <xs:attribute name="timeShiftBufferDepth" type="xs:duration" />
  <xs:attribute name="suggestedPresentationDelay" type="xs:duration" />
  <xs:attribute name="maxSegmentDuration" type="xs:duration" />
  <xs:attribute name="maxSubsegmentDuration" type="xs:duration" />
  <xs:anyAttribute namespace="##other" processContents="lax" />
</xs:complexType>
```

...
<!-- Period -->

DASH MPD Schema (3/5)

<!-- Period -->

```
<xs:complexType name="PeriodType">
  <xs:sequence>
    <xs:element name="BaseURL" type="BaseURLType" minOccurs="0" maxOccurs="unbounded"/>
    <xs:element name="SegmentBase" type="SegmentBaseType" minOccurs="0"/>
    <xs:element name="SegmentList" type="SegmentListType" minOccurs="0"/>
    <xs:element name="SegmentTemplate" type="SegmentTemplateType" minOccurs="0"/>
    <xs:element name="AdaptationSet" type="AdaptationSetType" minOccurs="0" maxOccurs="unbounded"/>
    <xs:element name="Subset" type="SubsetType" minOccurs="0" maxOccurs="unbounded"/>
    <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
  </xs:sequence>
  <xs:attribute ref="xlink:href"/>
  <xs:attribute ref="xlink:actuate" default="onRequest"/>
  <xs:attribute name="id" type="xs:string" />
  <xs:attribute name="start" type="xs:duration"/>
  <xs:attribute name="duration" type="xs:duration"/>
  <xs:attribute name="bitstreamSwitching" type="xs:boolean" default="false"/>
  <xs:anyAttribute namespace="##other" processContents="lax"/>
</xs:complexType>
```

<!-- Adaptation Set -->

DASH MPD Schema (4/5)

```
<!-- Adaptation Set -->
<xs:complexType name="AdaptationSetType">
  <xs:complexContent>
    <xs:extension base="RepresentationBaseType">
      <xs:sequence>
        <xs:element name="Accessibility" type="DescriptorType" minOccurs="0" maxOccurs="unbounded"/>
        <xs:element name="Role" type="DescriptorType" minOccurs="0" maxOccurs="unbounded"/>
        <xs:element name="Rating" type="DescriptorType" minOccurs="0" maxOccurs="unbounded"/>
        <xs:element name="Viewpoint" type="DescriptorType" minOccurs="0" maxOccurs="unbounded"/>
        <xs:element name="ContentComponent" type="ContentComponentType" minOccurs="0" maxOccurs="unbounded"/>
        <xs:element name="BaseURL" type="BaseURLType" minOccurs="0" maxOccurs="unbounded"/>
        <xs:element name="SegmentBase" type="SegmentBaseType" minOccurs="0"/>
        <xs:element name="SegmentList" type="SegmentListType" minOccurs="0"/>
        <xs:element name="SegmentTemplate" type="SegmentTemplateType" minOccurs="0"/>
        <xs:element name="Representation" type="RepresentationType" minOccurs="0" maxOccurs="unbounded"/>
      </xs:sequence>
      <xs:attribute ref="xlink:href"/>
      <xs:attribute ref="xlink:actuate" default="onRequest"/>
      <xs:attribute name="id" type="xs:unsignedInt"/>
      <xs:attribute name="group" type="xs:unsignedInt"/>
      <xs:attribute name="lang" type="xs:language"/>
      <xs:attribute name="contentType" type="xs:string"/>
      <xs:attribute name="par" type="RatioType"/>
      <xs:attribute name="minBandwidth" type="xs:unsignedInt"/>
      <xs:attribute name="maxBandwidth" type="xs:unsignedInt"/>
      <xs:attribute name="minWidth" type="xs:unsignedInt"/>
      <xs:attribute name="maxWidth" type="xs:unsignedInt"/>
      <xs:attribute name="minHeight" type="xs:unsignedInt"/>
      <xs:attribute name="maxHeight" type="xs:unsignedInt"/>
      <xs:attribute name="minFrameRate" type="FrameRateType"/>
      <xs:attribute name="maxFrameRate" type="FrameRateType"/>
      <xs:attribute name="segmentAlignment" type="ConditionalUintType" default="false"/>
      <xs:attribute name="subsegmentAlignment" type="ConditionalUintType" default="false"/>
      <xs:attribute name="subsegmentStartsWithSAP" type="SAPType" default="0"/>
      <xs:attribute name="bitstreamSwitching" type="xs:boolean"/>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<!-- Representation -->
```


DASH MPD Schema (5/5)

```
<!-- Representation -->
<xs:complexType name="RepresentationType">
  <xs:complexContent>
    <xs:extension base="RepresentationBaseType">
      <xs:sequence>
        <xs:element name="BaseURL" type="BaseURLType" minOccurs="0" maxOccurs="unbounded"/>
        <xs:element name="SubRepresentation" type="SubRepresentationType" minOccurs="0" maxOccurs="unbounded"/>
        <xs:element name="SegmentBase" type="SegmentBaseType" minOccurs="0"/>
        <xs:element name="SegmentList" type="SegmentListType" minOccurs="0"/>
        <xs:element name="SegmentTemplate" type="SegmentTemplateType" minOccurs="0"/>
      </xs:sequence>
      <xs:attribute name="id" type="StringNoWhitespaceType" use="required"/>
      <xs:attribute name="bandwidth" type="xs:unsignedInt" use="required"/>
      <xs:attribute name="qualityRanking" type="xs:unsignedInt"/>
      <xs:attribute name="dependencyId" type="StringVectorType"/>
      <xs:attribute name="mediaStreamStructureId" type="StringVectorType"/>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>

...

</xs:schema>
```

Segment Indexing

Segment Index in MPD only

```
<MPD>
```

```
...
```

```
<URL sourceURL="seg1.mp4"/>
```

```
<URL sourceURL="seg2.mp4"/>
```

```
</MPD>
```

seg1.mp4

seg2.mp4

...

```
<MPD>
```

```
...
```

```
<URL sourceURL="seg.mp4" range="0-499"/>
```

```
<URL sourceURL="seg.mp4" range="500-999"/>
```

```
</MPD>
```

seg.mp4

Segment Index in MPD + Segment

```
<MPD>
```

```
...
```

```
<Index sourceURL="sidx.mp4"/>
```

```
<URL sourceURL="seg.mp4"/>
```

```
</MPD>
```

sidx
.mp4

seg.mp4

Segment Index in Segment only

```
<MPD>
```

```
...
```

```
<BaseURL>seg.mp4</BaseURL>
```

```
</MPD>
```

sidx

seg.mp4

DASH Profiles

- ▶ Set of restrictions on the Media Presentation (MPD & Segments)
- ▶ Permission for DASH clients that only implement the required features
- ▶ 6 Profiles defined in ISO/IEC 23009:

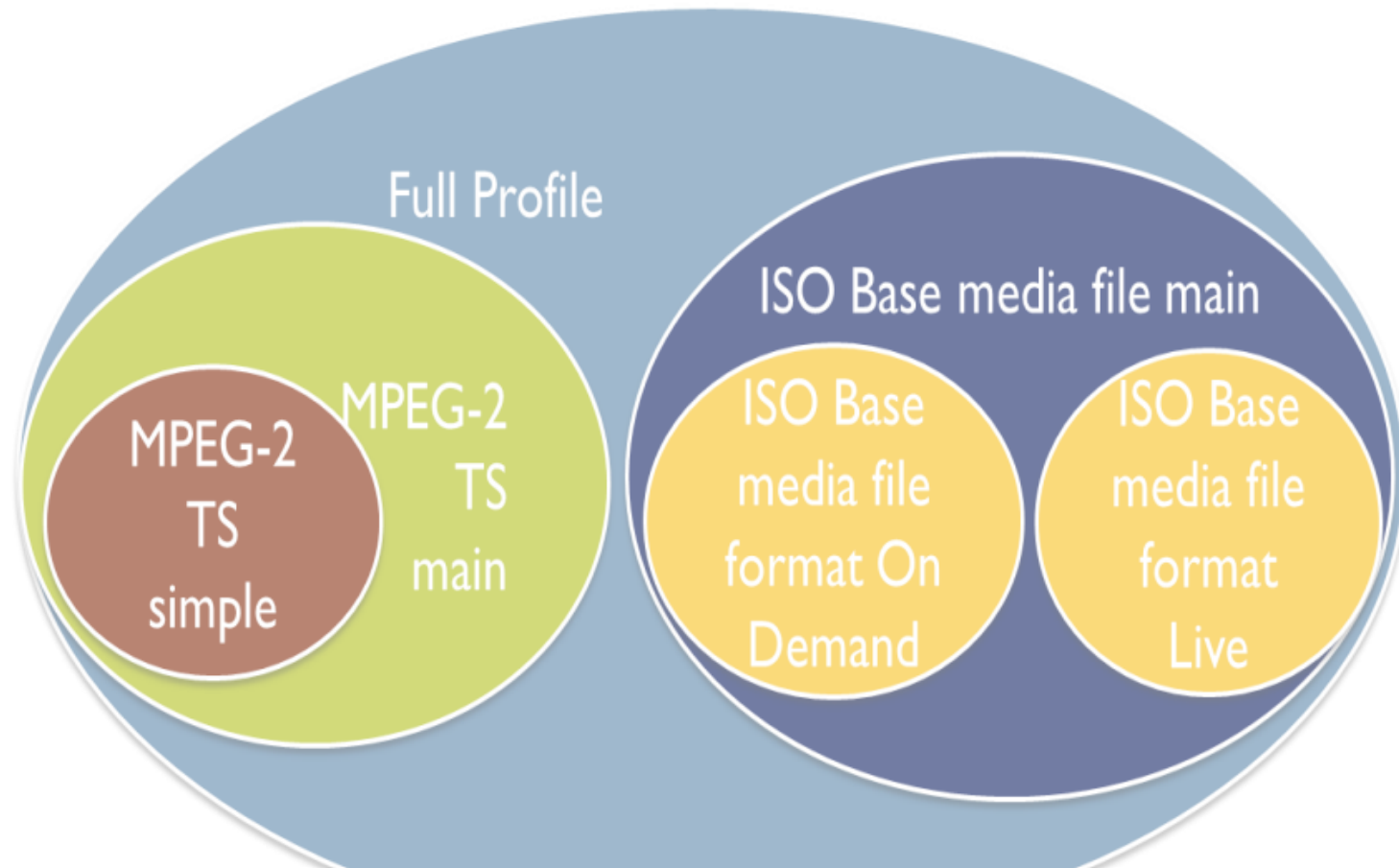


Table of contents

- Background
- Introduction
- Streaming
- Real time interactive application protocols
- Content distribution networks and multimedia contents
- New trends in MM protocols

Real time interactive application protocols

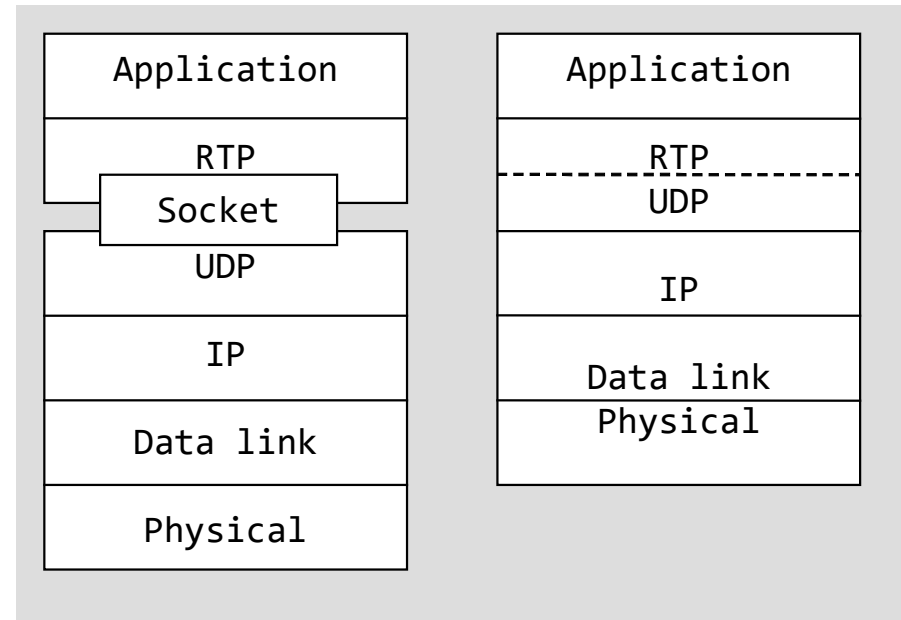
- RTP (Real Time Transport Protocol)
- RTCP (RTP Control Protocol)
- SIP (Session Initiation Protocol)
- H.323

Real Time Transport Protocol (RTP)

- Allows sending any audio (PCM, GSM, MP3) or video (MPEG, H.263) formats
- Complementary to other real time protocols, like SIP and H.323
- RTP encapsulates data chunks inside RTP packets, that travel inside UDP datagrams
- RTP does not assure that data arrives to destination, nor in order, as the protocol is only recognized end-to-end (not in the intermediate routers)

Real Time Transport Protocol (RTP)

- Associate different streams for audio and video, although they are combined and data is sent interlaced
- Part of the **application** layer, but it can also be seen as a sublayer inside **transport**



RTP Control Protocol (RTCP)

- Sends periodical control packets to the session participants over RTP
- Basic functions
 - Informs about distributed data quality
 - Maintains a persistent identifier for a RTP source. Allows recovering session participants
 - Controls sending rate of the RTP participants
 - Communicate session control information (i.e. to show user identification)

RTP Control Protocol (RTCP)

- Control packet types
 - **SR:** Sender report, sending and reception of active senders statistics
 - **RR:** Receiver report, sending and reception of non active sender statistics
 - **SDES:** Source description elements
 - **BYE:** Indicates session leave
 - **APP:** Specific application functionalities that the protocol uses

Session Initiation Protocol (SIP)

- Application level protocol that allows multimedia sessions establishment, modification and termination
- Some examples of use
 - Telephone calls through the Internet
 - Invite participants to existing sessions
 - Add and remove contents from a MM session
- User has a unique identifier, independently from his location (mobile phone, e-mail address)

Session Initiation Protocol (SIP)

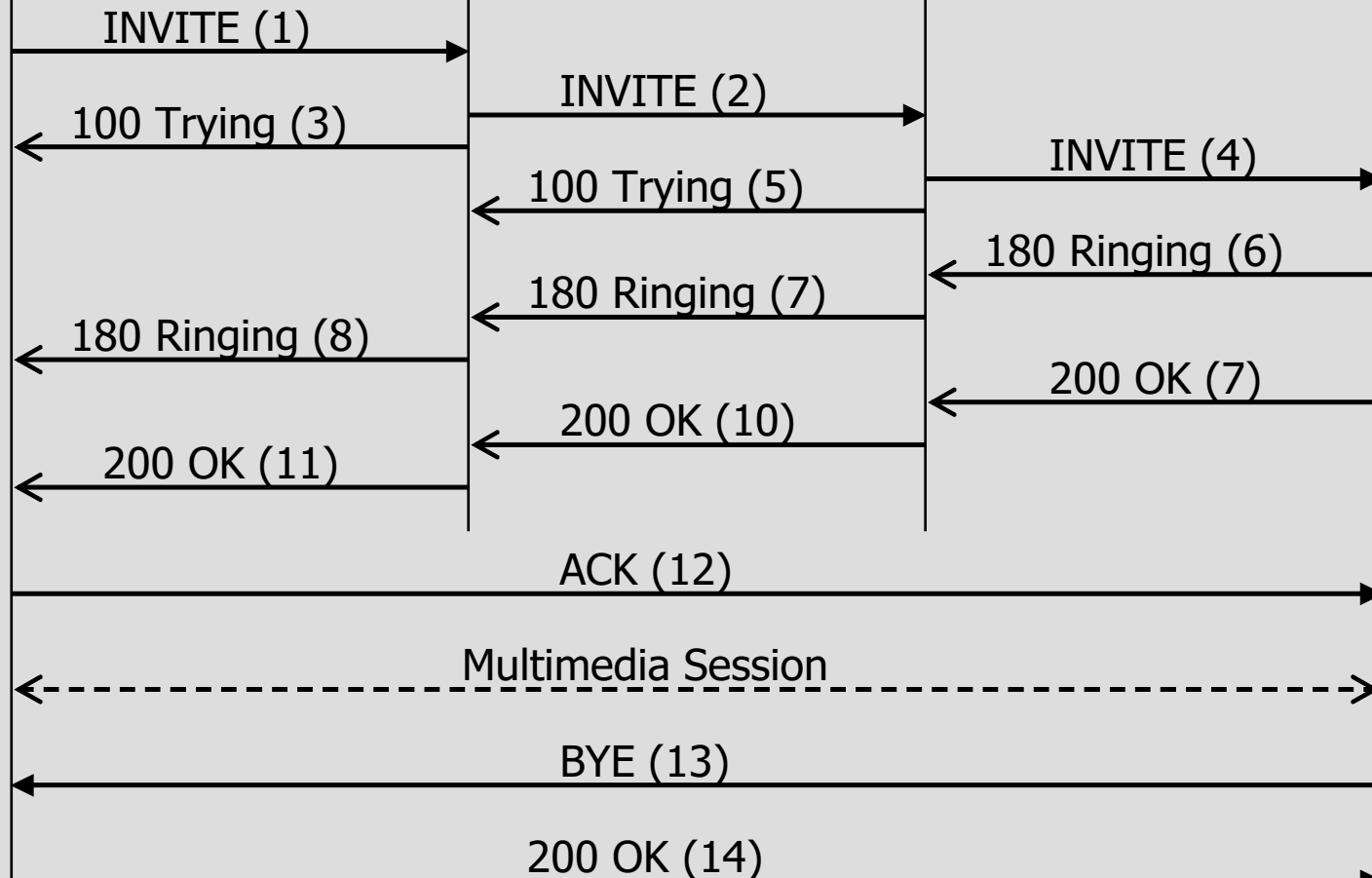
- Start / End connection
 - User location
 - User availability
 - User capabilities (media and parameters)
 - Start of the session: Call and establishment
 - Session management
- Not a complete communication system
 - Component that can be combined with other protocols
 - RTP, RTCP transport, RTSP streaming

Amalia's phone

Natalia's phone

bcn.cat

vic.cat



Session Initiation Protocol (SIP)

- SIP uses a response mechanism similar to HTTP
- To make a call:
 - Amalia calls Natalia using the SIP URI
 - Similar to an email address
 - Amalia uses a proxy server to locate Natalia
 - Sends an INVITE message
 - Uses SDP (Session Description Protocol) to send session data
- Sip addresses can be secure (sips)

Session Initiation Protocol (SIP)

- INVITE example

```
INVITE sip:amalia@bcn.cat SIP/2.0
Via: SIP/2.0/UDP pc01.bcn.cat;branch=z9hG4bK776asdhds
Max-Forwards: 70
To: Natalia <sip:natalia@vic.cat>
From: Amalia <sip:amalia@bcn.cat>;tag=1928301774
Call-ID: a84b4c76e66710@pc01.bcn.cat
CSeq: 314159 INVITE
Contact: <sip:amalia@pc01.bcn.cat>
Content-Type: application/sdp
Content-Length: 142

... Rest of SDP data ...
```

INVITE message content

- Method, source SIP URI and SIP version
- Rest of lines: Headers
 - From: Communication source
 - To: Communication destination
 - Via: Address where responses have to be received
 - Cseq: Sequence number that increases with each request
 - Contact: SIP URI that represents the direct route to the sender, user name + domain name
 - Max-Forwards: Maximum number of hops that the request can do
 - Content-type, content-length: Information about the message

INVITE response

SIP/2.0 200 OK

Via: SIP/2.0/UDP server10.vic.cat

;branch=z9hG4bKnashds8;received=192.0.2.3

Via: SIP/2.0/UDP bigbox3.site3.vic.cat

;branch=z9hG4bK77ef4c2312983.1;received=192.0.2.2

Via: SIP/2.0/UDP pc33.vic.cat

;branch=z9hG4bK776asdhs ;received=192.0.2.1

To: Natalia <sip:natalia@vic.cat>;tag=a6c85cf

From: Amalia <sip:amalia@bcn.cat>;tag=1928301774

Call-ID: a84b4c76e66710@pc33.vic.cat

CSeq: 314159 INVITE

Contact: <sip:natalia@192.0.2.4>

Content-Type: application/sdp

Content-Length: 131

... Rest of SDP data ...

INVITE response message content

- Response to the SIP request, response code
- Rest of lines: Headers
 - From: Communication source
 - To: Communication destination
 - Via: Address where we want to receive the responses
 - Cseq: Sequence number that increases with each request
 - Contact: SIP URI that represents the direct route to the receiver, user name + domain name
 - Content-type, content-length: Message information

Session Initiation Protocol (SIP)

- SIP address format
- sip:user:password@host:port;uri-parameters?headers
- Example: sip:amalia@bcn.cat
- For sips (secure sip), change sip by sips
- Rest of fields
 - user: Identifier of a resource in the host to which we are addressing. In this context, a host is a domain
 - Password: User password in clear (security risk!)

Session Initiation Protocol (SIP)

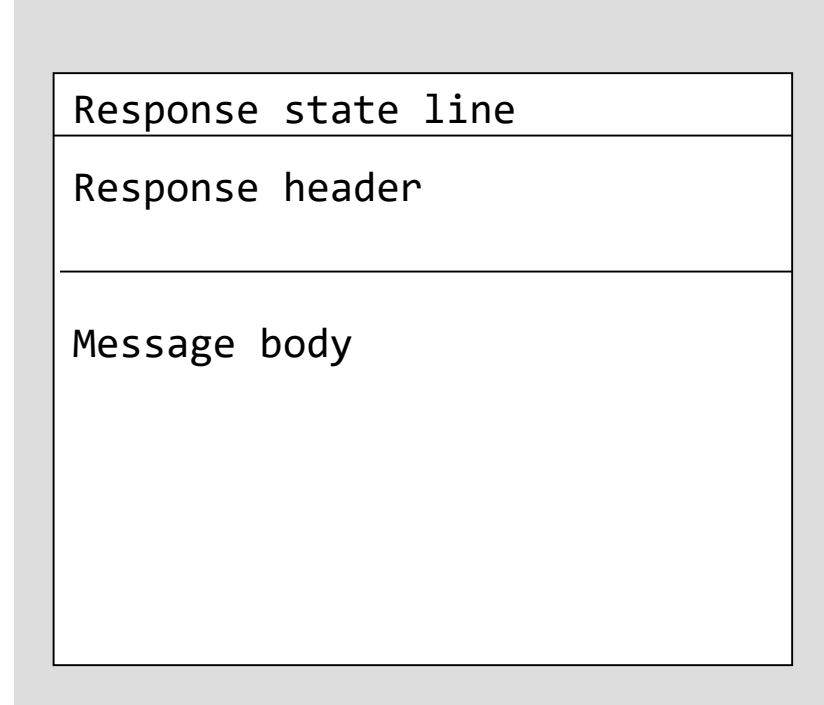
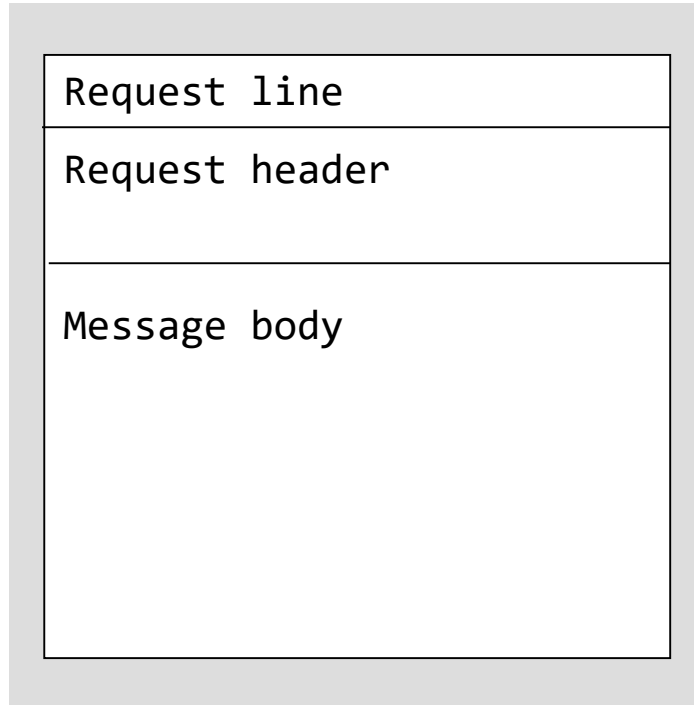
- Rest of fields
 - host: Host that offers the sip resource, can be a domain or an IP
 - port: Port number where the request is sent
 - uri parameters: Parameters affecting the requests carrying the URI. Represented as the HTTP parameters
 - headers: Header fields that can be included in the request represented by the URI

Session Initiation Protocol (SIP)

- SIP request message
 - Very similar to HTTP --> request line, headers, message body
- SIP methods functionality
 - **REGISTER**: Allows resource registration, to be able to make SIP communications afterwards
 - **INVITE**: Allows session initiation
 - **ACK**: Indicates that the request sent with INVITE has been accepted and the session has been established
 - **CANCEL**: Allows cancelling a request. An error message is sent
 - **BYE**: Allows ending the session
 - **OPTIONS**: Allows asking client or proxy server features

Session Initiation Protocol (SIP)

- SIP response message
 - Very similar to HTTP --> state line, headers, message body



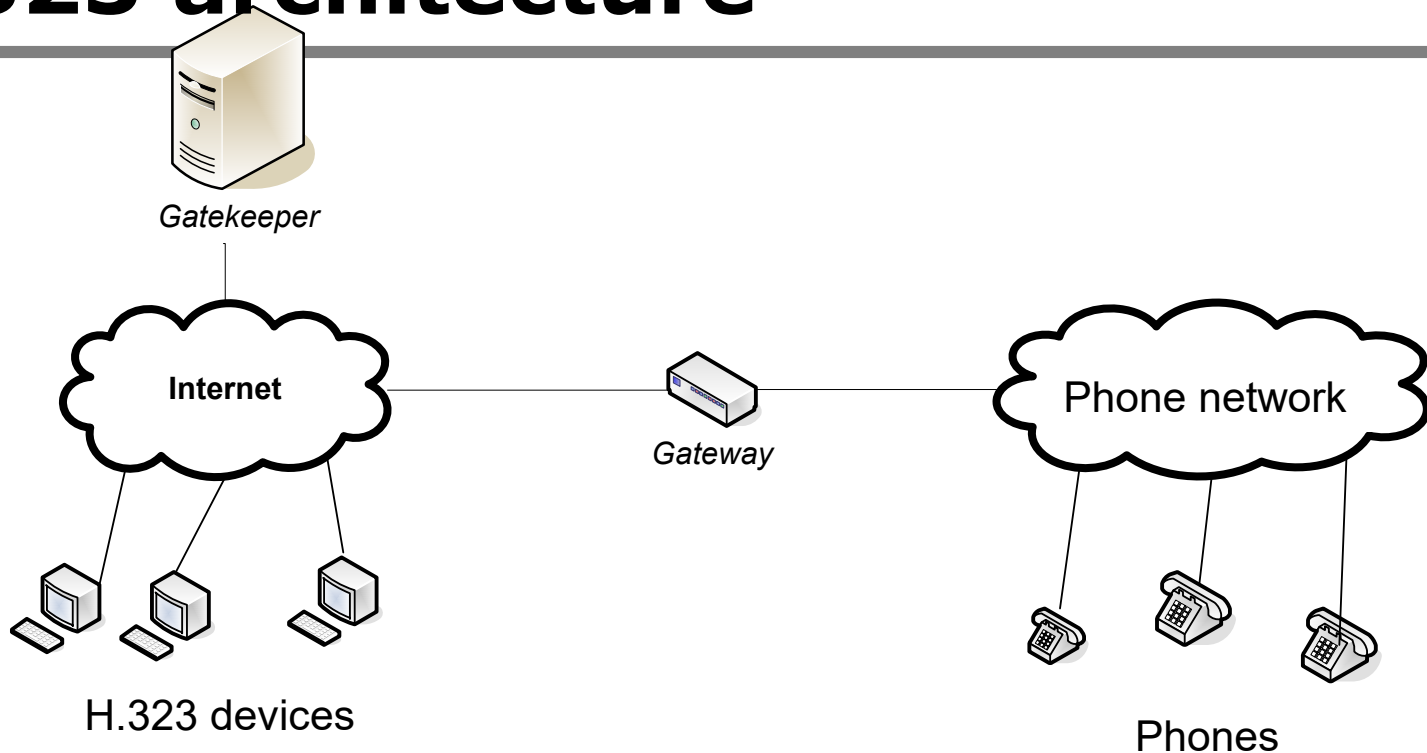
H.323

- Defined by ITU
- Alternative to SIP
- Allows audio and video transmission in real time
- It is used for VoIP communications and allows making calls to traditional telephone network devices
- Includes several specifications
 - End-to-end negotiation of audio and video codifications
 - RTP is usually used to send audio and video
 - Communication between devices and gatekeepers
 - Communication between devices connected to the telephone network and to the Internet

H.323

- Devices must support audio features. Video features are optional
- An example of architecture next

H.323 architecture



- Gateway: Optional element, it is used to interconnect devices between different kinds of networks
- Gatekeeper: Virtual switch, acts as a central point of calls in an area and offers several services to users

Reference: <https://www.geeksforgeeks.org/internet-telephony-protocol-h-323/>

Comparing H.323 and SIP

H.323	SIP
Integrate set of protocols for: signaling, register, admission control, transport and codecs	Start and management of the session, no transport nor audio/video formats supported
ITU (Telephony)	IETF (Internet)
Complex, several interconnected parts	Simple and easy to implement
Binary messages	ASCII, messages are longer and they do not work well in WAN networks
To send data, both use RTP/RTCP	
They need at least 3 ports, one for the session protocol and two more for RTP and RTCP	

SIP and H.323 usage

- Which application uses SIP / H.323?
 - https://support.zoom.com/hc/en/article?id=zm_kb&sysparm_article=KB0066041

Table of contents

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- Content distribution networks and multimedia contents
- New trends in MM protocols

Content Delivery Networks (CDN)

- Allow distributing the same content to many users
- Replicate content in several Internet locations to bring it closer to users and fasten its distribution
- Host servers in data centers to achieve higher speeds

CDN and stored MM

- When video is transmitted, user does not want to wait its complete download to watch it
- CDN charges provider for content transmitted
- How this issue can be solved?

CDN and stored MM

- When video is transmitted, user does not want to wait its complete download to watch it
- CDN charges provider for content transmitted
- How this issue can be solved?

CDN and stored MM

- Transmission and payment (option 1)
 - Streaming servers control user content download
 - CDN transmits data to the user
 - CDN charges provider for the complete information sent (even if the user does not watch it completely)
 - Problem for provider, it pays more than transmitted

CDN and stored MM

- Transmission and payment (option 1)
 - Streaming servers control user content download
 - CDN transmits data to the user
 - CDN charges provider for the complete information sent (even if the user does not watch it completely)
- Problem for provider, it pays more than transmitted

CDN and stored MM

- Payment (option 2)
 - Progressive download
 - Does not depend on a streaming server to provide the content
 - CDN transmits data to the user
 - CDN charges provider for what user downloads (not the complete content)
- Solves payment problem to provider

CDN and stored MM

- Payment (option 2)
 - Progressive download
 - Does not depend on a streaming server to provide the content
 - CDN transmits data to the user
 - CDN charges provider for what user downloads (not the complete content)
- Solves payment problem to provider

CDN and stored MM

- Data transmission
 - HTTP Streaming con Adaptive Bit Rate (ABR)
 - Base for MPEG-DASH
 - Several commercial products provide streams of different qualities that adapt to the user network speed
 - When user stops watching, transmission stops
 - Easy model to implement in a CDN
 - We can replicate content and send it as a classic web server
- Example: Amazon

CDN and Live MM

- CDN have evolved to support this kind of content transmission
- Support for several formats
- Problem when deciding which content should be cached
 - More demanded format
 - It does not mean extra storage in the CDN

References:

<https://www.dacast.com/blog/content-delivery-network-cdn-providers-live-streaming/>

<https://techdocs.akamai.com/msl/docs/create-your-media-services-live-streams>

-  Mapping System (centralized)
-  HL-DNS (centralized)
-  Monitoring Agent (distributed/located in Cluster)
-  LL-DNS (distributed/located in Cluster)
-  Edge Server (distributed, totally 119,000 servers)



- 1 Reporting Periodically (Monitoring Agent to Mapping System)**
 1. Health & Load of Clusters and Edge Servers (a)
 2. RTT & Packet Loss between Clusters and Local DNS Servers (b)
 3. RTT & Packet Loss between Akamai Clusters
- 2 Low Level Map Update: Every 2~10s (Mapping System to Low-Level DNS)**
 1. Edge Server Status in a Cluster: Health & Load of Edge Servers
 2. RTT & Packet Loss between Clusters and Local DNS Servers
- 3 High Level Map Update: Every 15~20m (Mapping System to High-Level DNS)**
 1. Mapping between LL-DNS Servers and Local DNS Servers
 2. Cluster Status: Health & Load of Cluster
 3. RTT & Packet Loss between Clusters and Local DNS Servers



ARCHITECTURE OF NETFLIX

N

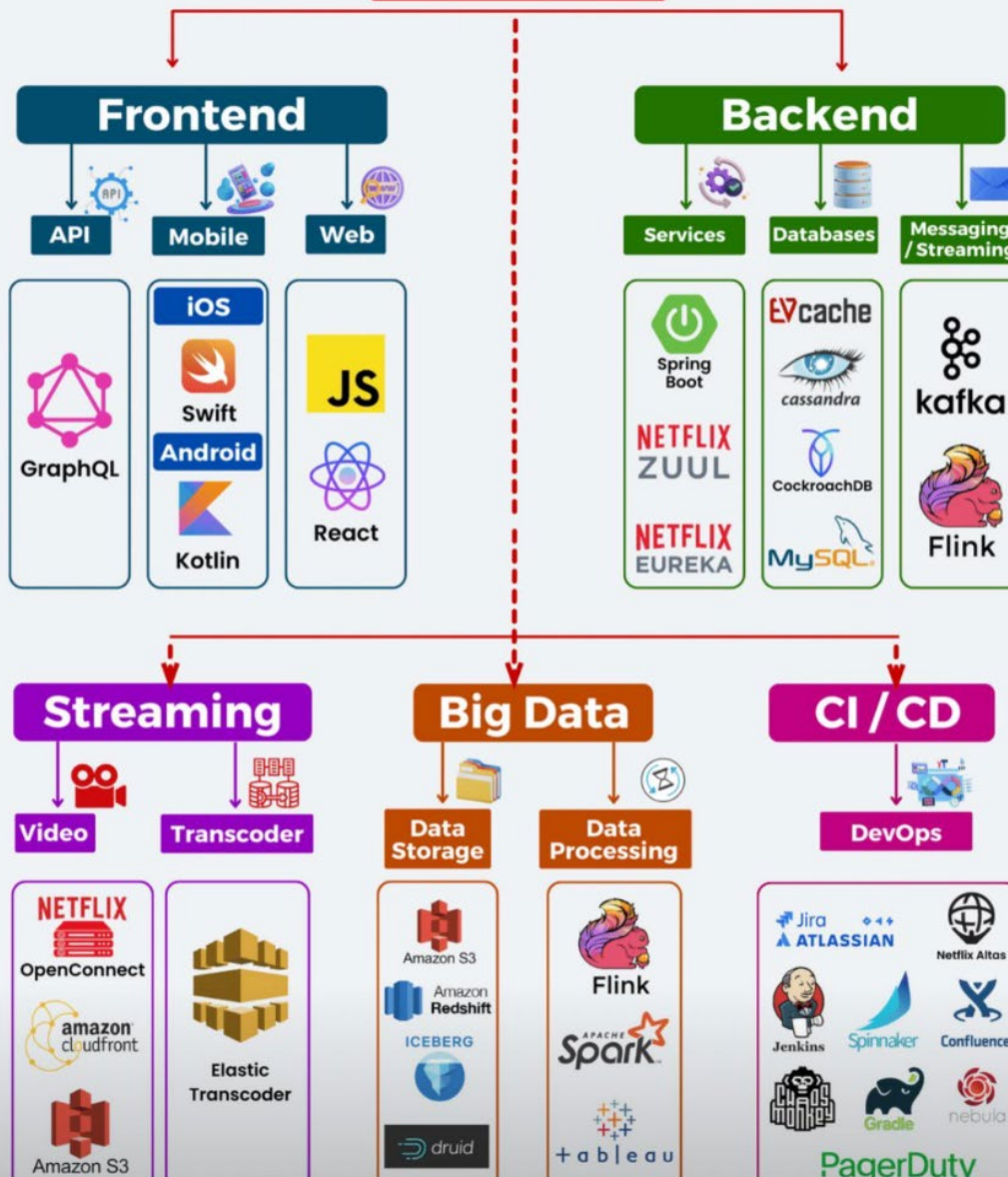


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WebRTC

- WebRTC 1.0: Real-Time Communication Between Browsers
 - W3C Recommendation, 8 October 2024
 - Combined with other application level protocols
 - Session Description Protocol (SDP)
 - Real Time Transport Protocol (RTP)
 - P2P (Peer-To-Peer) Communication Protocol between browsers

References:

<https://www.w3.org/TR/webrtc/>

<https://w3c.github.io/webrtc-pc/> (updated 30 October 2024!)

<https://antmedia.io/webrtc-transforms-live-streaming/>

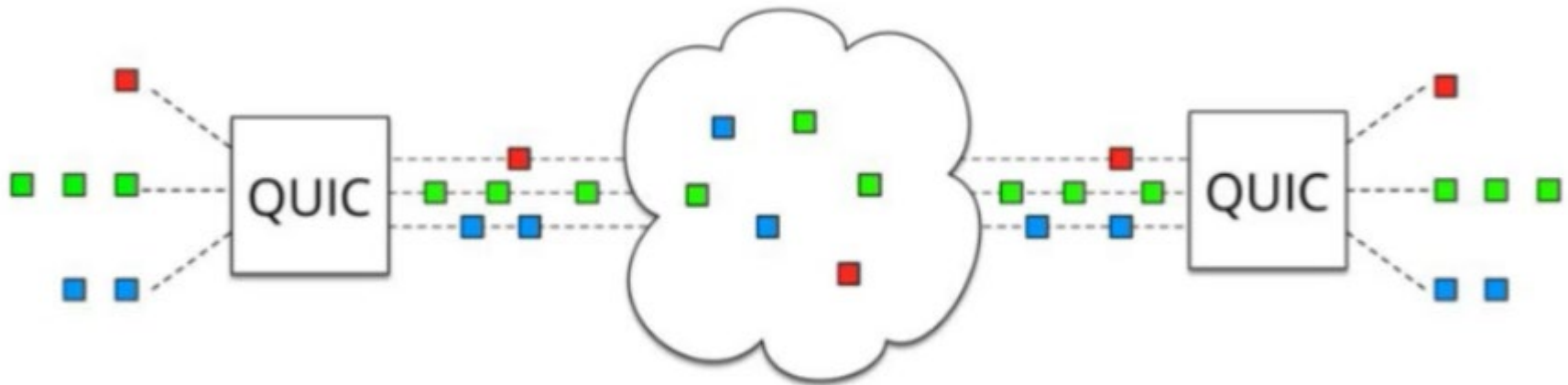
WebRTC, browser support

Peer To Peer		45
Connectivity		
WebRTC 1.0	Yes	✓
ObjectRTC API for WebRTC	Yes	✓
Data channel	Yes	✓
Input		
Access the webcam	Yes	✓
Screen Capture	Yes	✓
Enumerate devices	Yes	✓
Recording		
Media Stream recorder	Yes	✓

Curiosity: Comparison between Google duo and Google meet
<https://webrtcchacks.com/meet-vs-duo-2-faces-of-googles-webrtc/>

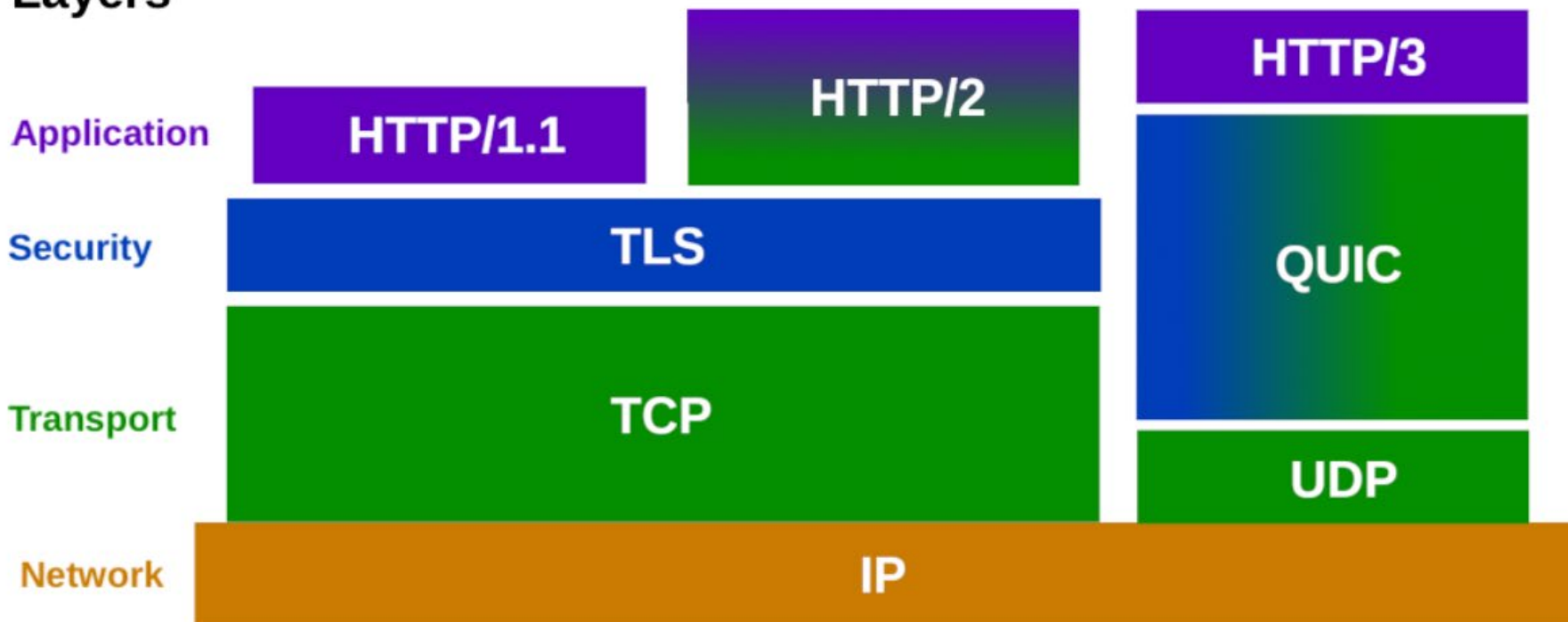
QUIC

- Web protocol developed over UDP
- Secure communication
- Faster connection establishment, three-way handshake not required → No connection at all
- With TCP, we are blocked when a packet is missing
 - Not with UDP



QUIC Architecture

Layers



www.humanlevel.com

<https://www.rfc-editor.org/rfc/rfc9114.html>

<https://datatracker.ietf.org/doc/html/rfc9114>

<https://www.humanlevel.com/blog/seo/http3-y-quic.html>

Testing HTTP3 support

- How can I know if a server supports HTTP/3?
- <https://domsignal.com/http3-test>
 - www.upc.edu
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Source: <https://kinsta.com/blog/http3/>

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- <https://github.com/MPEGGroup/DASHSchema>

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Multimedia content transmission and distribution systems

*Silvia Llorente **

DAC – UPC

* Part of the material comes from other sources. See bibliography